

The Energy Transition and Political Polarization Along Occupational Lines*

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Abstract

The green transition creates economic winners and losers. To what extent has the energy transition resulted in political polarization along occupational lines? We examine this question in the context of Germany, a country that has adopted some of the most far-reaching energy transition policies and that also has substantial employment in heavy emissions occupations. The far-right AfD party started campaigning as the only party opposing energy transition policies in 2016. Using a difference-in-differences design, we show that, after the 2013 election, AfD support increased in counties with larger shares of employment in polluting occupations. We then use individual-level panel data to show that since 2016, individuals in polluting occupations have become significantly more likely to identify with the far-right. We further demonstrate that these findings are not due to simultaneous political changes, most notably (i) differential activation of xenophobia after 2015 and (ii) losses in manufacturing employment. This suggests that far-right parties have benefited from concerns over present and future energy transition policies.

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1 Introduction

The green energy transition, a shift from nonrenewable to renewable energy sources worldwide, is causing wide-ranging changes in the labor market. There are new opportunities for “green” jobs related to environmental sustainability but also threats to those who have occupations that primarily exist in high-emissions industries; often labeled “brown jobs” (Vona et al., 2018)¹. Indeed, the energy transition poses an existential challenge for some industries and occupations that need to be “phased out”. This change may have adverse consequences. Workers in brown jobs often face difficulties when transitioning to occupations in other industries, and promises of financial compensation may not be credible (Bluedorn et al., 2023; Gazmararian and Tingley, 2023). A large-scale phasing out of brown jobs may also threaten communities that rely heavily on polluting industries and occupations. Do these labor market changes also have political consequences? In many countries, the far-right has been the first, the loudest, and often the only opponent of energy transition policies. Can the far-right attract support from workers in brown jobs and communities that depend on brown jobs?

We examine these questions in the context of Germany. Germany has long pursued relatively aggressive energy transition policies, which have affected polluting industries. The leader of Germany’s main industry association, Siegfried Russwurm, called the country’s energy policies “absolutely toxic” and warned that energy-intensive industries have moved away from Germany and will continue to do so.² Since 2016, the far-right Alternative for Germany (AfD) has become the only party that has vocally opposed these policies. The AfD emphasizes that energy transition policies pose existential threats to both economic freedom and to key German industries, especially the automotive, mining and chemical industries. We examine whether this has resulted in polarization along occupational lines, assessing changes in (i) voting behavior in regions with a larger presence of brown occupations, as well

¹We use terminology from prior work, which defines “brown jobs” as occupations which are relatively more prevalent in high-emissions industries

²“Olaf Scholz’s energy policies branded ‘toxic’ by German business chief” Financial Times, February 6, 2024.

as (ii) political preferences among individuals in those occupations.

We follow the literature in defining a brown occupation as an occupation that is much more likely to be in a highly polluting industry than in any other industry (Cavallotti et al., 2024; Vona et al., 2018). Individuals in these occupations are likely to be more invested in the economic success of polluting industries than people who work in a polluting industry but whose occupation is more general, such as someone who works in marketing or financial administration. To measure aggregate-level variation in brown employment, we obtain administrative labor market data on the share of workers in these occupations. We hypothesize that the energy transition, and the ensuing existential threat to brown occupations, may affect (i) communities that depend on these jobs and (ii) the individuals who work in brown occupations.

Our first set of results draws on a county-level panel of federal election results between 2005 and 2021. Using supplementary survey data, we first demonstrate that the AfD party adopted a consistent and wide-ranging stance opposing the energy transition in 2016. Since this shift should differentially affect regions with a higher share of employees in brown occupations, it allows us to estimate a difference-in-difference specification, comparing elections before and after 2016. We find that, compared to the 2013 election, AfD support increased in counties with high shares of brown employment in 2017 and 2021. By contrast, the Greens lost in counties with higher brown employment shares, particularly in 2021. Importantly, trends in far-right support up until 2013 did not vary with the prevalence of employment in brown occupations. We further show that electoral shifts after 2013 cannot be attributed to a correlation between brown employment and (i) manufacturing employment, (ii) pre-existing levels of xenophobia as well as support for anti-immigrant parties and (iii) urban-rural differences.

In addition to the aggregate-level results, we rely on a large-scale annual panel survey to analyze changes in political preferences among individuals employed in brown occupations. Similar to the county-level results, we find that employees in brown occupations dispropor-

tionately shifted towards the far-right AfD party beginning in 2016. Even though some workers leave these occupations during this time period, the increase in AfD support among workers in brown occupations cannot be explained purely by compositional changes among this group. Notably, we further discuss that individual-level changes in AfD support are not sufficient to explain our aggregate-level results, since workers in brown occupations are a relatively small group. This suggests that the energy transition can increase far-right support also among individuals who live in areas that are more reliant on brown jobs, even if those individuals do not work in directly affected occupations.

These findings contribute to an emerging literature that examines new domestic political cleavages over the energy transition. As the distributive consequences of the energy transition become more apparent, politicians have started to court those who have lost or who expect to lose when their industries are phased out. In many countries, far-right parties have been the most vocal and often the only opponent of energy transition policies. This suggests that climate may become a third pillar of far-right support beyond opposition to immigration and globalization (Colantone et al., 2022; Voeten, 2024). Most of the existing literature has focused on political opposition to energy taxes or other measures that affect the cost of carbon intensive consumption. We focus on the labor market implications of the energy transition, which thus far has only received attention in regard to specific sectors of the economy (Bolet, Green and González-Eguino, 2023; Egli, Schmid and Schmidt, 2022; Gazmararian, 2024b; Goetz et al., 2019). In contrast, our study provides evidence on all occupations that are affected by the energy transition. In addition, we present new results based on both a panel of aggregate-level voting, as well as based on within-individual variation in political preferences over time.

2 Theory and Literature

2.1 The Labor Market and Electoral Consequences of the Energy Transition

Scholars have attributed the rise of the far-right in Europe and the United States to both cultural and economic factors. The political economy literature has identified three plausible economic drivers of support for radical right parties and candidates: deindustrialization due to globalization and automation, austerity following the 2008 financial crisis, and the economic consequences of migration (For recent overviews of the literature, see Colantone and Stanig (2019); Margalit (2019); Rodrik (2021)).

An emerging literature focuses on the energy transition as a fourth potential economic driver of support for far-right parties and candidates. Most of this literature highlights how energy transition policies increase the costs of carbon intensive consumption, which may drive some voters towards parties and candidates that oppose climate policies and away from Green parties or others who take strong pro-climate positions. For example, Milanese voters who were adversely affected by a ban on polluting cars became more likely to vote for the Lega Nord (Colantone et al., 2022). Similarly, a Dutch policy that increased household energy taxes and redistributed the revenues as subsidies for renewable energy led voters who were more exposed to increases in household energy taxes and who did not benefit from the subsidies to become more likely to vote for the radical right and less likely to support the Greens (Voeten, 2024). These findings build on a prior literature showing that voters are sensitive to gasoline price increases (e.g. Kim and Yang, 2021) and a public opinion literature demonstrating that public support for climate policies depends on their cost to individuals and their redistributive effects (e.g. Bechtel and Liesch, 2020; Beiser-McGrath and Bernauer, 2023; Bernauer and Gampfer, 2015; Egan and Mullin, 2017; Schaffer, 2023; Stokes and Warshaw, 2017).

Yet, the energy transition and decarbonization will also have substantial labor market im-

plications. Energy transition policies rely on making carbon intensive production relatively more expensive, for example through emission trading schemes, carbon taxes, regulations, or subsidies on less carbon intensive production. Most European countries have adopted environmental taxes and regulations that have increased energy prices. There is substantial evidence that higher energy prices have reduced labor demand in energy intensive industries across Europe in the 2010s (Bijnens, Konings and Vanormelingen, 2022; Cox et al., 2014; Marin and Vona, 2021). Moreover, sociologists have started to examine the increased stigmatization associated with polluting industries and occupations (Beckfield and Evrard, 2023), suggesting the possibility that efforts to fight climate change reduce the social status of brown occupations.

What sets the energy transition apart is that the threat to brown industries and occupations is existential: some industries and jobs will disappear altogether if governments follow through on their climate commitments (Colgan, Green and Hale, 2021). Green and other parties that advocate for more aggressive energy transition policies explicitly call for a phase out of fossil fuels, especially coal. Such policy positions have drastic consequences for individuals with brown occupations and for communities that either produce fossil fuels or have industries that heavily rely on fossil fuels, such as the chemical industry. While the energy transition also creates new greener jobs, individuals rarely move from more carbon-intensive to greener jobs (Bluedorn et al., 2023). Promises of financial compensation or green jobs that might replace brown jobs may not be credible (Gazmararian and Tingley, 2023).

In many European countries, climate policy has long been a near consensus issue among centrist parties. Green parties have advocated for more aggressive energy transition policies, but there was little active opposition to existing policies until far-right parties started to take on the issue in the mid-2010s (Huber et al., 2021). In a paper close to our study, Cavallotti et al. (2024) show that workers who benefit economically from the energy transition exhibit support for the energy transition and parties advocating for it. Using cross-sectional survey data in 15 countries and constructing similar “greenness scores”, Cavallotti et al. (2024) find

that people in greener jobs are more likely to support environmental policy, Green parties, and parties with pro-environmental policy platforms, while the opposite holds for people with browner jobs. It is thus plausible that the (anticipated) labor market consequences of decarbonization may have effects on support for the radical right.

There is a literature on the electoral implications of the decline in coal employment in the United States. The decline of coal in the U.S. is mostly a consequence of increased competition from natural gas rather than climate policies. Yet, parties and politicians have taken visibly different positions about the future of coal. Especially Donald Trump campaigned heavily to defend coal whereas the Obama administration was accused of waging a “war on coal.”³ Districts with a higher share of coal-related job losses have relatively higher shares of the Trump vote (Egli, Schmid and Schmidt, 2022; Goetz et al., 2019) and there is evidence that the drop in coal employment has led to a partisan shift towards Republicans in U.S. coal communities (Gazmararian, 2024b). There is also evidence that, beyond coal, individuals employed in high emission industries and occupations are less likely to support climate cooperation (Bechtel, Genovese and Scheve, 2019). However, we know less about whether the energy transition contributes to broader electoral polarization along occupational lines.

Our study builds on the substantial literature that attributes the rise of the radical right in the United States and Europe at least in part to job losses in import competing industries (e.g. Autor et al., 2020; Colantone and Stanig, 2018; Dippel et al., 2022). Most manufacturing job losses are likely due to automation (Baldwin, 2019) but increased trade exposure to especially China has also contributed to the decline of manufacturing, which long provided stable and relatively well compensated employment for a less educated male dominated labor force. In the United Kingdom, the anti-EU wing of the Conservative party and far-right politicians, such as Nigel Farage, managed to attract votes in communities adversely affected by import competition (Colantone and Stanig, 2018). In Germany, the far-right was similarly successful in communities and among individual voters affected by

³Grunwald, Michael (1 June 2014). “New Carbon Rules the Next Step in Obama’s War on Coal”. *Time*. Retrieved 6 January 2016.

job losses from import competition (Dippel et al., 2022).

A key theoretical precondition for these economic effects to have electoral effects is that candidates or parties take distinct positions on them. For example, in the U.S., Donald Trump departed from longstanding Republican support for trade. Scholars therefore identify the effect of manufacturing employment losses on changes in Republican vote shares between 2012, when Romney ran on a traditional Republican platform, and 2016, although the labor market implications of the China shock were already visible in 2012 (Autor et al., 2020). Identifying when the issue becomes salient and polarized is thus a prerequisite for any study of the electoral effects of the energy transition.

Theories of how globalization affected support for the far-right through its effects on labor markets differ across two key conceptual dimensions. First, some studies highlight the labor market effects on individuals who are directly affected by deindustrialization whereas others also focus on the effects on communities (e.g. Broz, Frieden and Weymouth, 2021). Second, some studies examine purely economic pathways, such as job or wage losses or changes in housing prices, while others focus on how economic shocks affect the social status and values of individuals and groups (e.g. Baccini and Weymouth, 2021; Ballard-Rosa, Jensen and Scheve, 2021). The broadest studies argue that industrial decline not only affect the political views of affected worker, but they can also have broad economic and social effects on communities (Autor, Dorn and Hanson, 2019; Baccini and Weymouth, 2021; Pierce and Schott, 2020).

We take a similarly broad approach. It is theoretically plausible that the energy transition affects electoral choices because individuals in brown occupations fear that their job opportunities or their social status are adversely affected. It is also plausible that the transition has broader effects on communities. For example, there is evidence that the energy transition creates considerable existential uncertainty in fossil fuel communities (Gazmararian, 2024a; Gazmararian and Tingley, 2023) and that individuals tend to prefer broad based community approaches to compensation for job losses (Gaikwad, Genovese and Tingley,

2022). Sociologists have also identified the broader social implications of threats to relatively well-paying jobs for a relatively less-educated and male dominated labor force (for an overview, see: Beckfield and Evrard, 2023). It is thus entirely plausible that concerns about energy transition policies have both individual and community level effects and that they can work through economic and/or social channels.

2.2 The German Case

Germany is a good case to study this question because it has a relatively advanced energy transition policy and a relatively large share of the population working in polluting industries. The 2000 Renewable Energy Act adopted a feed-in tariff and grid priority for renewables. In 2010 the CDU/CSU-led German government formally endorsed the “Energiewende,” a planned transition towards a low carbon (and nuclear-free) economy. These policies were supported by various coalition governments led by the centrist Christian Democrats (CDU/CSU) joined by the right of center FDP and the left of center Social Democrats (SPD).

The policies helped make Germany a leader in renewable energy industries (Nahm, 2017). However, Germany also remains a traditional industrial powerhouse. For example, about 20 percent of Germany’s workforce is in manufacturing, compared to just 10 percent for the United States and 12 percent in France. The car manufacturing industry is only just beginning to transition, resulting in significant job losses as electric vehicles are easier to build. The chemical industry cannot easily transition away from fossil fuel combustion. For example, BASF, which has more than 50,000 employees in Germany, has announced that it will permanently cut operations in Germany in response to “high energy prices and overregulation.”⁴

There is evidence that higher electricity prices have resulted in lower demand for labor in German industries with high electricity demand, and that the effect is greater among low-skilled workers (Cox et al., 2014). A European study found that the regions with the greatest

⁴<https://www.reuters.com/markets/europe/basf-says-european-operations-need-be-cut-size-permanently-2022-10-26/>

potential negative labor impact of rising electricity prices are located in Belgium, Germany, the Netherlands, Northern Italy, and Sweden (Bijnens, Konings and Vanormelingen, 2022). We do not yet know what, if any, the political consequences of these labor market changes are. Evidence from Spain suggests that an effective, just transition policy that offers generous compensation to coal miners and coal mining communities gave an electoral boost to the party that implemented a coal phase-out (Bolet, Green and González-Eguino, 2023). Survey experimental evidence from the United States also supports this general insight (Gazmararian, 2024a). This matters in our context, as Germany has quite generous compensation policies for individuals who lose employment, including retraining programs. For example, in 2020, the German parliament passed the Structural Strengthening Act for Coal Regions (“Strukturstärkungsgesetz Kohleregionen”) which aims to mitigate the impact of phasing out coal power by providing up to 40 billion euros until 2038 for significant investments and various support measures in coal regions, including active labor market policies, promoting their development post-coal era.⁵

For job concerns to translate into voting behavior, there must be salient distinctions between parties or candidates. Other than the AfD, German parties have not campaigned on climate change opposition during our time frame. We determine the timing of the AfD’s shift to climate opposition using three data sources: party manifestos, media reports, and election surveys. The AfD party program for the 2013 Federal elections remained silent on climate change.⁶ The 2014 program for the European elections claims that scientific evidence for anthropogenic climate change is uncertain. However, the program still supports international climate mitigation efforts, claiming that: “In order to take the precautionary principle into account, a gradual reduction in CO2 emissions can be agreed within the framework of international agreements.”⁷

⁵<https://www.bmwk.de/Redaktion/DE/Textsammlungen/Wirtschaft/strukturstuerkungsgesetz-kohleregionen.html>.

⁶https://www.abgeordnetenwatch.de/sites/default/files/election-program-files/afd_1.pdf

⁷<https://www.abgeordnetenwatch.de/sites/default/files/election-program-files/afd-europawahl-2014.pdf>

This changes completely with the AfD’s 2016 party Grundsatz (“basic principles”) program, which attacks both climate science and climate policies. The document claims that “Carbon dioxide (CO²) is not a pollutant, but an indispensable component of all life.” The program also tackles the Energiewende and its existential threat to industry and Germany ways of life (see also figure 1). The program states that: “[...] the German government is abusing the increasing CO₂ concentration for the “great transformation” of society, with the result that personal and economic freedom is massively restricted.”⁸ Since then, the AfD has only increased its attention to this issue. In 2019, AfD leading politician Alexander Gauland declared the opposition to the “so called climate change policies” to be among the three most important topics for the AfD⁹. Its 2021 program again challenges climate science and the “radical restructuring of industry and society” that “threatens our freedom on an increasingly frightening scale.”¹⁰ Importantly, the AfD focused on price increases and social injustices related to the energy transition, especially on the question of ‘who pays?’ for the transition (the AfD’s answer being ‘the core people’) (Küppers, 2024). The AfD has also consistently voted against climate legislation in the Bundestag and the European Parliament, where it was one of only three national party groups to vote against all major climate proposals (Schaller and Carius, 2019).¹¹ There is also recent anecdotal evidence that the AfD gained substantially in opinion polls over vocal opposition to a law mandating a transition to electric heatpumps.¹²

The media search examines whether the AfD’s shift was noticed in public debates. We did a Factiva search on articles on the topic climate change that also mention the AfD for Germany’s four largest newspapers: Bild, Süddeutsche Zeitung, Franfurter Allgemeine, and Die Welt. This yields 997 articles, but only one that appeared before 2016. The 2016 articles

⁸<https://www.afd.de/wp-content/uploads/2016/03/Leitantrag-Grundsatzprogramm-AfD.pdf>

⁹Welt am Sonntag, 28.08.2019, <https://www.welt.de/politik/deutschland/article201093000/C02-Emissionen-Die-AfD-und-die-sogenannte-Klimaschutzpolitik.html>

¹⁰https://www.afd.de/wp-content/uploads/2021/06/20210611_AfD_Programm_2021.pdf

¹¹The others were the PVV (The Netherlands) and UKIP (the United Kingdom).

¹²Karl Mathiesen, “How the far-right weaponized heat pumps into electoral rocket fuel” Politico, October 4 2023, <https://www.politico.eu/article/robert-lambrou-alternative-for-germany-heat-pump-election-climate-change/>

Figure 1: AfD election poster



Notes: English translation: “Stop the EEG [(*German Renewable Energy Sources Act*)] and the energy transition. Electricity should not be a luxury!” Picture taken in September 2017 in Leverkusen. Retrieved from Internet Initiative Leverkusen e.V.

focus on the AfD’s new election program, which was discussed in all newspapers and invited numerous critical opinion pieces. The jobs angle is also apparent. For example, newspaper stories report that AfD top candidate Alice Weidel in Berlin accuses Chancellor Angela Merkel (CDU) of endangering jobs in the automotive industry and unnecessarily pursuing the “de-industrialization of Germany.”¹³ By far the most stories on the AfD and climate appear in 2019. Most of these relate to the Friday for future protests and reflect a broader jump in attention for the climate issue in Germany that year. More generally, the salience of climate issues increased considerably. We did a Factiva search on German media on the subjects “climate” and “elections”. In 2013, this subset includes only 86 stories, but in 2017 it has 1358 stories and in 2021 9008. There are 226 stories in 2016, despite the absence of Federal elections that year. This suggests that the issue became a salient electoral issue only after 2015.

Data from election surveys confirm the timing of climate change polarization. Figure 2

¹³“AfD bestreitet Klimawandel”, *Süddeutsche Zeitung*, 5 September 2017

shows how voters for different parties place themselves on the climate change issue based on the German Longitudinal Election Study (GLES). The figure plots regression coefficients and their 95 percent confidence intervals relative to CDU/CSU voters. In 2013, AfD voters are not significantly different from CDU and FDP voters. In contrast, in 2017 and even more so in 2021, the AfD voters stand out in terms of their views that the government should prioritize the economy over climate change (in 2017)¹⁴ and that the government has already done too much to fight climate change.¹⁵ The AfD is the only party that is substantively and statistically significantly to the right of the mainstream center right parties on this issue.¹⁶

The AfD changed their position on the energy transition in 2016 primarily based on strategic considerations. In part, this stemmed from a change in the party leadership in 2015. This leadership change increased the party’s tendency to oppose policies endorsed by the government and all mainstream parties (Küppers, 2024; Spoon, Hobolt and de Vries, 2014). In particular, the AfD aimed to differentiate itself from parties like the Greens by opposing the energy transition and by promoting a broader political narrative seeking to return to a “good old” industrial era. Moreover, party professionalization and spillovers from other radical right parties who succeeded by running on climate-skeptic positions also contributed to this policy shift. Importantly, secondary evidence on the AfD platform shift does not suggest that this shift was driven by changes among AfD supporters or potential supporters. We provide more details in section A.1 in the SI.

Having identified 2016 as the first year the AfD supplied a clear policy platform that opposed the energy transition, we thus test the following hypotheses:

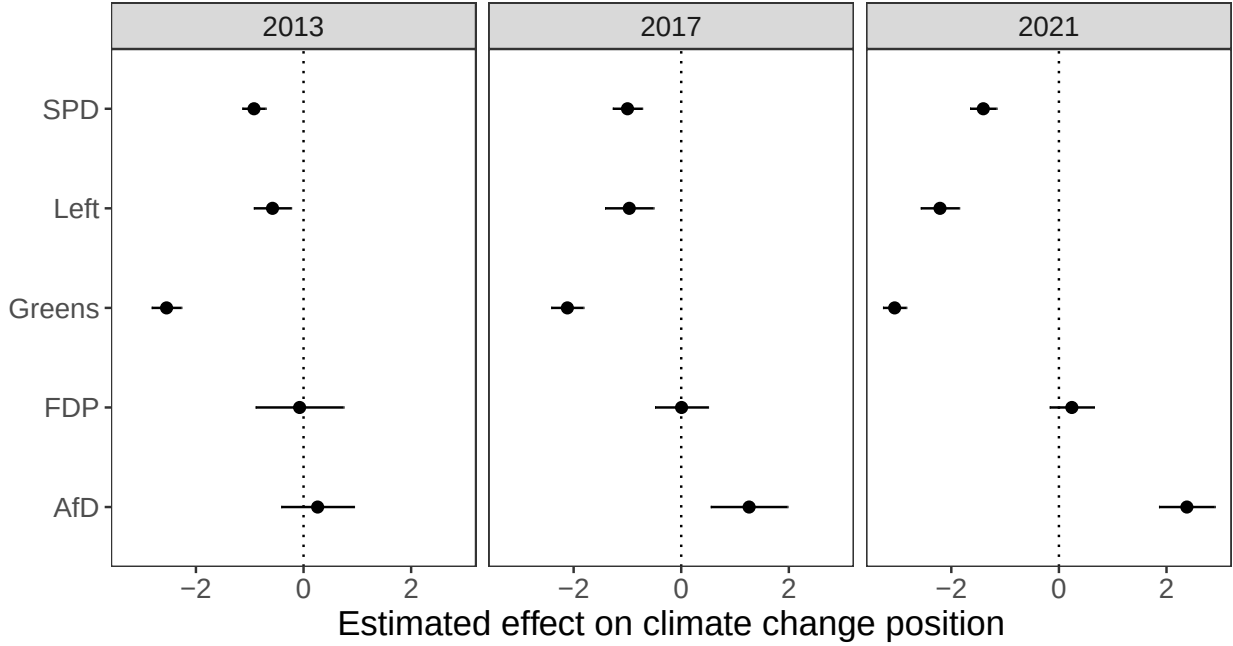
- **H1:** Communities with a larger share of voters in brown occupations become more likely to vote for the far-right in 2017 and 2021 compared to 2013.

¹⁴The 2017 question asks: “Should combating climate change be a priority in any case, even if it harms economic growth”

¹⁵The 2021 question is: “Some think that much more needs to be done in politics to combat climate change. Others think that politics to combat climate change have already gone way too far.”

¹⁶Voters were also asked to place parties on this scale, although they were not asked about the AfD in 2013. In 2017 and especially 2021, voters recognized the AfD as being significantly “to the right” on climate from the other parties.

Figure 2: Voter views on climate change by party choice over time



Notes: The figure shows linear regression coefficients and 95% confidence intervals relative to CDU/CSU voters (using survey weights). The question wording in 2013 and 2017 was: “Should combating climate change be a priority in any case, even if it harms economic growth, or should economic growth be a priority in any case, even if it makes combating climate change more difficult?” In 2021 the question was: “Some think that much more needs to be done in politics to combat climate change. Others think that politics to combat climate change have already gone way too far.”

- **H2:** Individuals in brown occupations become more likely to identify with the far-right after 2015.

3 Data

Our analysis draws on (i) aggregate-level voting data and (ii) an individual-level panel that records both occupation and party preferences. Our aggregate-level data links administrative employment data and federal election results at the county level for elections between 2005 and 2021. At the individual level, we use the German Socio-Economic Panel (SOEP), a longitudinal survey of households established in 1984 and carried out annually.

3.1 Brown occupations

Our analyses require us to distinguish employees in brown- and non-brown occupations. To do so, we rely on brownness scores for occupations provided by Vona et al. (2018). Vona et al. (2018) first define the most polluting industries as those in the 95th percentile of pollution intensity for at least three out of eight relevant pollutants (CO₂, CO, VOC, NO_x, SO₂, PM₁₀, PM_{2.5}, and lead). Then, Vona et al. (2018) define brown occupations as occupations which are relatively more prevalent in these polluting industries. Specifically, this means that an occupation is classified as “brown” if its prevalence in polluting industries at least 7 times higher than its prevalence in any other industry.

Importantly, the original classification by Vona et al. (2018) derives from the American context. Therefore, we use crosswalks provided by Cavallotti et al. (2024) to adapt them to the European context. This mapping results in a continuous measure of the “Brownness” of all occupations in the ISCO-08 (International Standard Classification of Occupations) classification. This measure ranges from 0 to 1 and represents the occupation-specific likelihood of being employed in a polluting sector. Because this classification is based on the US labor market, we assume that industry-specific pollution and occupational distribution across industries in the US resemble those in Germany. We believe that this assumption is justified, since production technologies and industrial trends are similar in the US and Germany (Baily, Bosworth and Doshi, 2020). We further probe the robustness of our results to an alternative measure based on German emissions and employment data developed by Südekum and Rademacher (2024). In particular, we use a variable that measures the share of full-time employees who work in industries with CO₂ emission increases between 2000 and 2019 for each county in the year 2019 (see A.2.1 for more details). Results presented in figure A.10 and discussed in section 6 are robust to this alternative treatment specification.

The brownness-scores allow us to distinguish employees in brown- and non-brown jobs for all ISCO occupations. To then create county-level measures of brown employment, we have obtained administrative labor market data from the German Federal Employment Agency

(*Bundesagentur für Arbeit*). This data measures the share of workers who are employed in these brown occupations in each county for the period 1996-2022. In all county-level analyses, our main treatment is measured as the share of people working in brown occupations in 2013, i.e. prior to the AfD platform change.

$$D_c = \frac{\sum_{a=1}^A N_{ac} \times \mathbf{1}(\text{brownness}_a > 0.3)}{\sum_{a=1}^A N_{ac}} \quad (1)$$

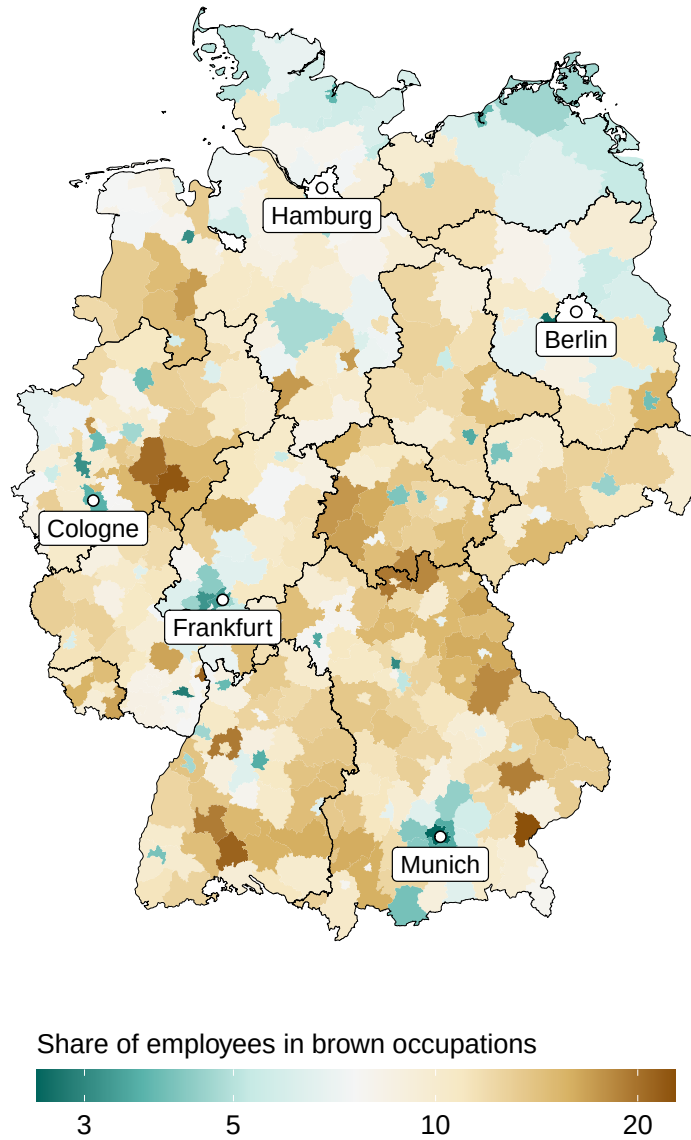
where N_{ac} measures the number of people working in a particular occupation a (ISCO-08) in a county c in 2013 and brownness_a measures the particular brownness score of that occupation. Following Cavallotti et al. (2024), we code occupations as brown if their brownness score is greater than 0.3. This threshold is guided by an economic interpretation. Jobs above the threshold can be considered as brown (e.g. sheet-metal workers, metal working machine tool setters), whereas jobs with scores below 0.3 are more ambiguous (e.g. butchers, fishmongers, food preparers). As we discuss in section 6, our main results are not dependent on this threshold – we find similar results when choosing alternative thresholds and an alternative treatment that does not dichotomize the brownness score.

We present the distribution of brown jobs across counties in 2013 in figure 3. High-emission jobs cluster in the highly industrialized regions in Baden-Württemberg, North Rhine-Westphalia, and Thuringia, and in the German chemical hubs, such as in Ludwigshafen which hosts the headquarters of chemical company BASF, and Altötting, which is at the center of the “Southeast-Bavarian chemical triangle”. We further list the most common high-emission jobs in the SOEP panel data in table A.1 in the SI. This set of jobs includes machine makers and operators in the agricultural, chemical, and manufacturing industries.

Individual-level data: When analyzing individual-level panel data, we obtain individual-level brownness scores using the ISCO-88 and ISCO-08 indicators in the SOEP panel data.¹⁷

¹⁷ISCO-88 codes are only available until 2017, ISCO-08 only since 2013. We use ISCO-08 codes to merge with brownness scores since 2013. For previous years, we cross-walk the ISCO-88 indicator with

Figure 3: Geographic distribution of brown jobs in Germany in 2013



Notes: The figure shows the share of brown jobs in each county in 2013. To better convey geographic variation, we use a log scale.

As with the county-level analysis, we create a binary indicator for brown occupation (brownness > 0.3). For each employed survey respondent, this indicator measures whether the

the ISCO-08 codes using the official correspondence table of the International Labor Organization (<https://www.ilo.org/public/english/bureau/stat/isco/docs/corrtab08-88.xls>). In some cases, multiple ISCO-08 codes are merged into a respective ISCO-88 code. In such cases, we calculate the mean brownness score across all the merged ISCO-08 codes.

respondent works in a relatively more “brown” occupation. The binary treatment indicator distinguishes the individual-level analysis from the county-level analysis. In the county-level analysis, we use the *share* of employees for which brownness > 0.3 .

3.2 Political outcomes

Aggregate-level outcomes: For the county-level analyses, we compile data on German federal election results at the county level between 2005 and 2021. Some counties experience border changes and mergers during this period. To obtain a balanced panel of comparable counties, we map all counties that experience boundary changes to their 2021 boundaries using population-weighted crosswalk files provided by the Federal Institute for Research on Building, Urban Affairs, and Spatial Development (BBSR; see section A.2.2 for more details).

Individual-level outcomes: When analyzing individual-level panel data, we focus on a survey item that measures individuals’ party preferences. In particular, participants are asked “Which party do you lean toward?”. We construct binary variables for each party that take the value 1 for a respective party if the respondents indicate that they lean toward that party, and 0 otherwise.

3.3 Covariates

At the county level, we collect data for several covariates. In particular, we collect data on the total county population and population density obtained from BBSR. We collect data on the proportion of manufacturing workers (as a percentage of the working population) from the Federal Statistical Office of Germany. Information on average household income and is derived from the national economic accounts of the federal states (Volkswirtschaftliche Gesamtrechnung der Länder). We also assess county-level GDP per capita data, which are available from the Annual Regional Database of the European Commission’s Directorate General for Regional and Urban Policy (ARDECO). Finally, we obtain data on

unemployment rates and the share of people with a higher education entrance qualification (Hochschulreife) from Inkar.

Table 1: Summary statistics for county-level variables in 2013

	Mean	Median	SD	N	Min	Max
Employees in brown occupations (%)	10.07	10.12	3.90	399	2.56	22.68
Unemployment rate (%)	6.43	5.90	2.99	399	1.30	15.20
GDP per capita	32633.70	28759.56	14394.74	399	14316.03	127642.62
Household income	20159.53	20035.00	2455.12	397	15186.00	33310.00
Share manufacturing (%)	19.79	18.58	8.53	397	1.65	49.38
Population density	509.45	193.61	657.45	399	36.47	4530.48
Population	193.85	149.70	165.81	399	34.10	1746.30
Share of high school graduates	31.88	30.04	10.24	399	0.00	59.11
Share of foreigners	7.09	6.22	4.60	399	0.87	31.27
AfD	489.72	472.71	111.39	399	237.69	900.89
Far right parties	655.11	616.26	180.46	399	292.59	1339.31
FDP	434.77	434.09	134.55	399	152.39	1041.60
CDU/CSU	4236.29	4194.84	734.86	399	2518.43	6258.88
SPD	2544.02	2441.93	737.95	399	1138.73	4986.06
Green Party	756.95	728.17	311.15	399	245.33	2173.36
Left Party	890.81	570.09	714.78	399	257.86	2944.71

Notes: The table shows summary statistics for aggregate-level data in 2013.

4 Research design

The goal of our empirical strategy is to assess changes in party support for (i) counties with higher shares of employees in brown occupations and (ii) individuals employed in these occupations. In a causal inference framework, the “treatment” can be thought of as a change in the party position of the AfD, which may differentially affect (i) regions with more employees in brown occupations and (ii) individuals who work in these occupations. Since the party platform change occurred in 2016, the 2013 election can be considered the last pre-treatment period for the aggregate-level analysis. Similarly, we can use 2015 as the last pre-treatment analysis for the panel survey respondents. Then, we can implement a typical difference-in-differences design. Here, identifying the treatment effect of the AfD

platform shift derives from the assumption that counties / individuals that are more affected by the AfD platform shift would have followed the same electoral trends as those that are not affected by the platform shift, had the platform shift not occurred.

4.1 County-level data

To estimate the aggregate effects of the energy transition on voting, we estimate a specification of the following form:

$$Y_{cjt} = \gamma_c + \delta_{jt} + \sum_{k=-3}^1 \theta_k (D_c \times \mathbf{1}_{t=k}) + \sum_{k=-3}^1 \mu_k (M_c \times \mathbf{1}_{t=k}) + \sum_{k=-3}^1 \pi_k (P_c \times \mathbf{1}_{t=k}) + \beta' X_{ct} + \varepsilon_{cjt} \quad (2)$$

Here, c indexes counties, t indexes elections and j indexes administrative districts.¹⁸ The outcome Y_{cjt} is the vote share of a given party in county c and election t . As discussed above, the treatment D_c is the share of people working in brown occupations in county c in 2013 (that is, to the AfD platform shift). We are interested in the parameters θ_k , which are the difference-in-differences estimates for the k^{th} election before or after treatment. Here, k indexes elections relative to 2017, with $k = 0$ indicating the 2017 election, $k = -1$ indicating the 2013 election, etc. The term γ_i is a county fixed effect, and the term δ_{jt} is an administrative district-specific election fixed effect. The latter accounts for time-varying confounding at the level of the administrative district (*Regierungsbezirk*) level. As is standard practice in event study specifications, we omit the interaction between D_c and the last pre-treatment period election (2013, which corresponds to $k = -1$). The 2013 election serves as the baseline for all estimated treatment effects. Standard errors are clustered by county.

The vector X_{ct} contains the following control variables, all measured at the county level: average household income, county-level GDP/capita, the share of workers employed in the

¹⁸Administrative districts (*Regierungsbezirke*) are a NUTS-2 administrative level below the state (NUTS-1), but above the county level (NUTS-3). However, only four populous states contain more than one administrative district: Bavaria, Hesse, Northrhine-Westphalia, and Baden-Württemberg. For all other states, there is only one administrative district per state. As we discuss in section 6, our conclusions remain unchanged when we use state-election fixed effects or election fixed effects that do not vary across higher administrative units.

manufacturing sector, population density, total population, the share of people with a higher education entrance qualification, the share of foreigners and the unemployment rate. We also include terms that interact the share of employees in manufacturing firms M_c as well as population density P_c with election dummies. These terms serve to account for the fact that brown jobs and manufacturing jobs may overlap, and that brown jobs are more prevalent in rural areas. In essence, the $M_c \times \mathbf{1}_{t=k}$ term means that we allow for separate trends in electoral outcomes, comparing counties with larger or smaller shares of employees in manufacturing. Similarly, the $P_c \times \mathbf{1}_{t=k}$ term allows for differential trends when comparing urban and rural areas.

4.2 Individual-level data

In a second step, we assess whether (i) party preferences differ between individuals in brown and non-brown occupations and (ii) whether this relationship varies over time. To do so, we estimate a specification of the following form:

$$y_{it} = \gamma_i + \delta_t + \sum_{k=2010}^{2021} \theta_k (D_{it} \times I_{t=k}) + \beta' X_{it} + \varepsilon_{it} \quad (3)$$

Here, y_{it} is the reported party preference for individual i in year t . Our main independent variable is D_{it} , the indicator of having a brown job. We include individual fixed effects γ_i and year-fixed effects δ_t . The vector of control variables X_{it} includes age, education levels, income, migration background, and concerns about immigration. We interact the brown job indicator D_{it} with indicator variables $I_{t=k}$. This specification is similar to a standard event study specification, but differs with respect to the term D_{it} . In a typical event study specification, the treatment group D_{it} is constant across t . In our setting, the specification allows for over-time variation in whether individuals are employed in a brown job, and therefore means that the different coefficients θ_k are *not* estimated using the same groups of people for each year t . This variation primarily stems from individuals who switch out of

brown occupations.

To assess whether compositional differences account for the over-time changes in political preferences of employees in brown occupations, we rely on two strategies: first, we include several time-varying covariates X_{it} , chiefly income and worries about immigration. We note that individual-level time-invariant characteristics, such as gender, education, and migration background, are already taken into account by including individual-level fixed effects γ_i . Second, we present additional specification that include weights. We use this weighting approach to keep the composition of the groups defined by D_{it} constant. We provide more details on this approach in section A.3 of the SI, where we also assess the resulting improvement in balance through a comparison of covariate averages over time in figure A.2. Although some imbalance remains, this approach significantly improves balance between the brown and non-brown groups over time. As we discuss in section 6, our main results are robust to the inclusion of these weights.

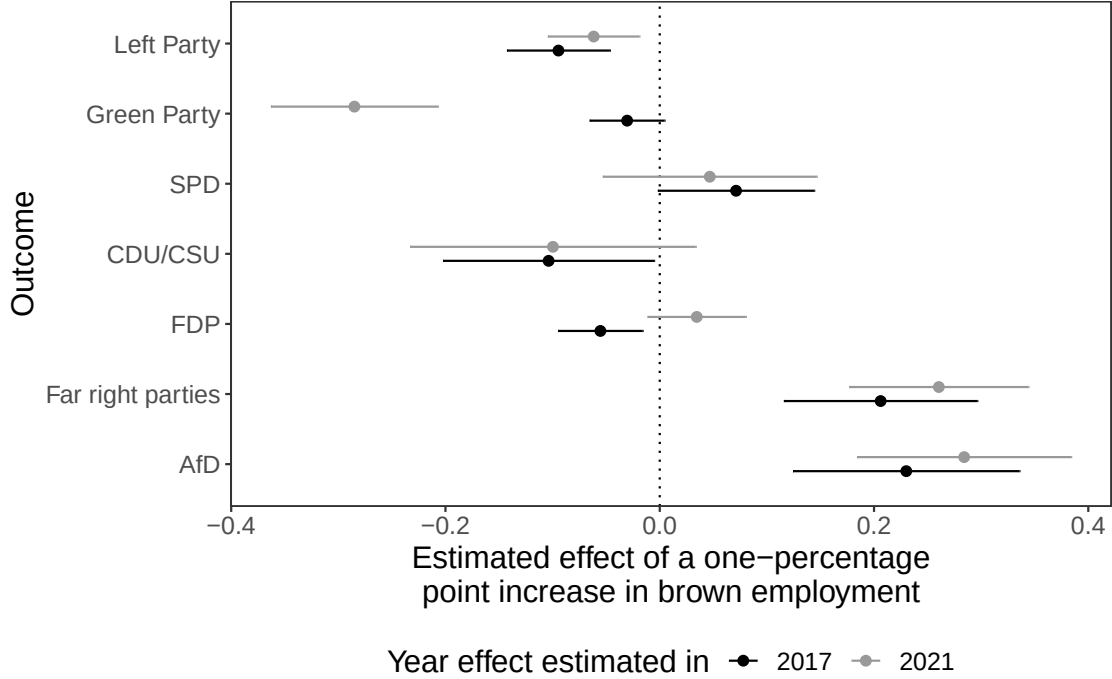
5 Results

5.1 Aggregate-level results

We begin by assessing whether counties with a higher share of brown jobs have experienced a change in party support after 2015. We present the results in figure 4. The figure shows the relationship between the share of brown jobs in 2013 and the change in vote shares between 2013–2017, as well as 2013–2021, at the county level. This means that we present the estimated coefficients θ_0 and θ_1 from equation 2.

In 2017, we find that counties with a higher share of brown jobs have experienced an increase in support for the AfD and a decrease in support for the Left party, Greens, and the FPD. In 2021, these patterns are similar, with the main differences being that losses among individuals in brown occupations for the Green party are considerably larger, while gains for the AfD are comparable to 2017. The 2021 elections similarly saw losses for the Left Party,

Figure 4: Aggregate results



Notes: The figure presents the results from the main specification. In particular, we present the coefficients θ_0 (for 2017) and θ_1 (for 2021) from the specification (equation 2) in section 4.1. Covariates are listed in section 4.1. Standard errors are clustered at the county level. The “far-right parties” term is the sum of all far-right parties, which includes that AfD. For details on the results, see table A.3.

whereas losses for the FDP are not observed in 2021. The estimates we present in figure 4 are consistent with a visual inspection of the relationship between brown employment shares and far-right support, which we present in the form of scatter plots in figure 5 as well as figure A.3 in the SI. The simple bivariate evidence in these figures demonstrates a clear positive relationship between increases in AfD voting after 2013 and levels of brown employment in 2013.

In the appendix, we further present lags and leads estimates in figure A.5. Here, we show all estimated coefficients θ_k from equation 2. As is standard in difference-in-differences designs, we use the estimated coefficients for the leads of the treatment to assess the validity of the parallel trends assumption. Regarding our main outcome, voting for the AfD party, we cannot directly assess trends prior to 2013, since the AfD did not compete in earlier elections. However, we can use the combined vote share of far-right parties instead, since

Figure 5: Change in AfD support 2013–2021 and brown occupations in 2013



Notes: The figure shows the county-level shares of brown occupations in 2013 and the change in AfD support between 2013 and 2021. We highlight select counties.

other far-right parties competed before 2013 (see section A.2.4 for more information on these parties). Here, we observe coefficients that are close to zero, suggesting that, with respect to radical-right voting, counties with many brown jobs were not on a different trend prior to 2013. We do, observe some evidence for violations of the parallel trends assumption prior to 2013, in particular for the Green Party, SPD and FDP (but not for the AfD or far-right outcomes). However, the magnitude of the difference-in-differences estimates prior to 2013 for these parties is relatively small.

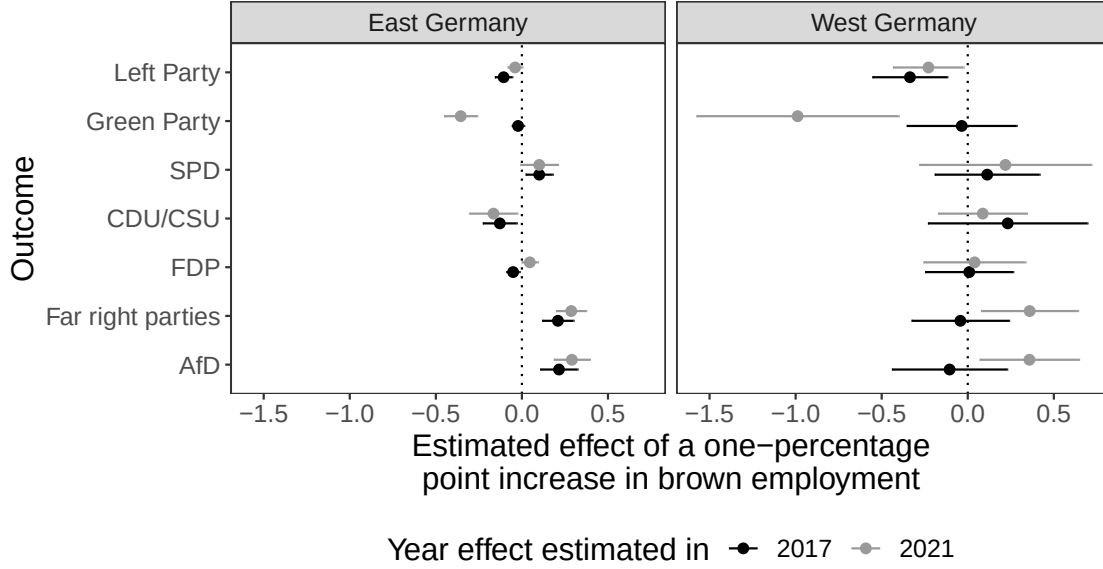
Sensitivity to possible parallel trends violations: To assess the severity of potential parallel trends violations, we implement an additional robustness check based on Rambachan and Roth (2023). In essence, this test allows us to assess whether our main results hold,

assuming that post-2013 parallel trends violations are as severe as the ones observed prior to 2013. We implement this test in figure A.9. With respect to our main outcome of interest, voting for the AfD, we find that a parallel trends violation as severe as the one observed in figure A.5 is *not* sufficient to completely explain the observed positive effect on far-right vote shares. Relatedly, we also note that a potentially more robust specification using the Sun and Abraham (2021) estimator results in considerably less evidence for parallel trends violations than the main specification (see figure A.6).

East-West differences: In the German context, a typical source of heterogeneity is the East-West comparison. In figure 6, we present the results from our main specification, subsetting the sample to either only East or only West Germany. We find that the observed results in both samples are generally comparable – the AfD gains and the Green Party loses, particularly in 2021. The main difference is that positive effects on AfD support are only observed for 2021 in West Germany, while they are observed for both 2017 and 2021 in East Germany. We note that the results in figure 6 further demonstrate that our results are not driven by larger increases in far-right support in East Germany. This implication is also visually confirmed in figure 5, where we observe a positive relationship between brown job prevalence and increases in AfD support for the 2021 election in *both* East and West Germany.

Changes in income and unemployment: In table A.5, we investigate whether counties with higher shares of brown employees experience changes in household income, GDP/capita and unemployment rates after 2013. We do not find robust evidence for changes in either household income or unemployment rates. We do observe that, relative to counties with lower brown employment shares, GDP/capita is declining. In 2021, our best estimate of the magnitude of this effect is that moving from the 25th to the 75th percentile of brown employment shares in 2013 is associated with declines of 2,930 Euros in GDP/capita in 2021. This is equal to about nine percent of average county-level GDP/capita in 2013.

Figure 6: Aggregate results – comparing East and West Germany



Notes: The figure presents the results from the main specification, separately for East and West Germany. In particular, we present the coefficients θ_0 (for 2017) and θ_1 (for 2021) from the specification (equation 2) in section 4.1. Covariates are listed in section 4.1. Standard errors are clustered at the county level. The “far-right parties” term is the sum of all far-right parties, which includes that AfD.

5.1.1 Alternative explanations

Before moving on, we assess whether our aggregate-level measure picks up other explanations for geographic variation in changes in AfD support after 2013.

First, an alternative economic hypothesis for the rise of the far-right is the (trade-induced) loss of manufacturing jobs (Dippel et al., 2022). Our main specification already accounts for this by interacting the share of employees in manufacturing with election dummies. In figure A.12, we further present results based on adding a term that interacts *changes* in manufacturing employment with election indicators. This specification allows for differential trends in AfD support that are correlated with local-level shifts in manufacturing. We find that our main results remain unchanged after accounting for differential trends due to changes in manufacturing employment between 2013 and 2017.

Second, another alternative explanation is that aggregate brown employment shares may be correlated with anti-immigrant sentiments or xenophobia. Since one dominant political

topic after 2013 was the inflow of refugees starting in 2015, our main results may reflect the activation of xenophobia in regions with higher brown employment. To assess whether this is the case, we obtain data on pre-existing levels of xenophobia. First, we measure the number of anti-refugee violence incidents and demonstrations in each county in 2015 obtained from Benček and Strasheim (2016) (see appendix A.2.3 for a description of the data). In our analysis, we use two versions of this measure – one scaled by population and one scaled by the total number of crimes in county c in 2015. We then re-estimate our main specification, allowing trends in party vote shares to vary by pre-existing levels of anti-refugee violence and demonstrations. Second, we further interact election indicators with 2013 levels of far-right support, which should similarly proxy for anti-immigrant sentiments. In figure A.13, we again find that the main effect of brown employment is robust to these additional tests. This implies that our main effects unlikely to be driven by more negative responses to the refugee inflow in areas where brown occupations are more prevalent.

Third, we can exclude the possibility that urban-rural differences drive our main results. As discussed in section 4.1, our main specification already allows for differential trends in vote shares that are correlated with population density. This should account for the fact that brown jobs are more prevalent in rural areas.

Finally, it is worth noting that the inclusion of various controls as well as region-specific time trends does decrease the magnitude of the estimated main coefficients (see table A.4 in the SI). For 2017, the magnitude of the coefficient for the far-right voting outcome is about 28.5% lower in specifications with covariates and region-specific time trends than in the base specification that only includes the interaction between elections and 2015 brown employment levels. For 2021, the magnitude decreases by about 32.1%. This suggests that there is some confounding due to, for example, variation in manufacturing employment, xenophobia, or urban-rural differences that are correlated with brown employment shares. However, this confounding is not sufficient to fully explain our main effects in figure 4 and table A.3.

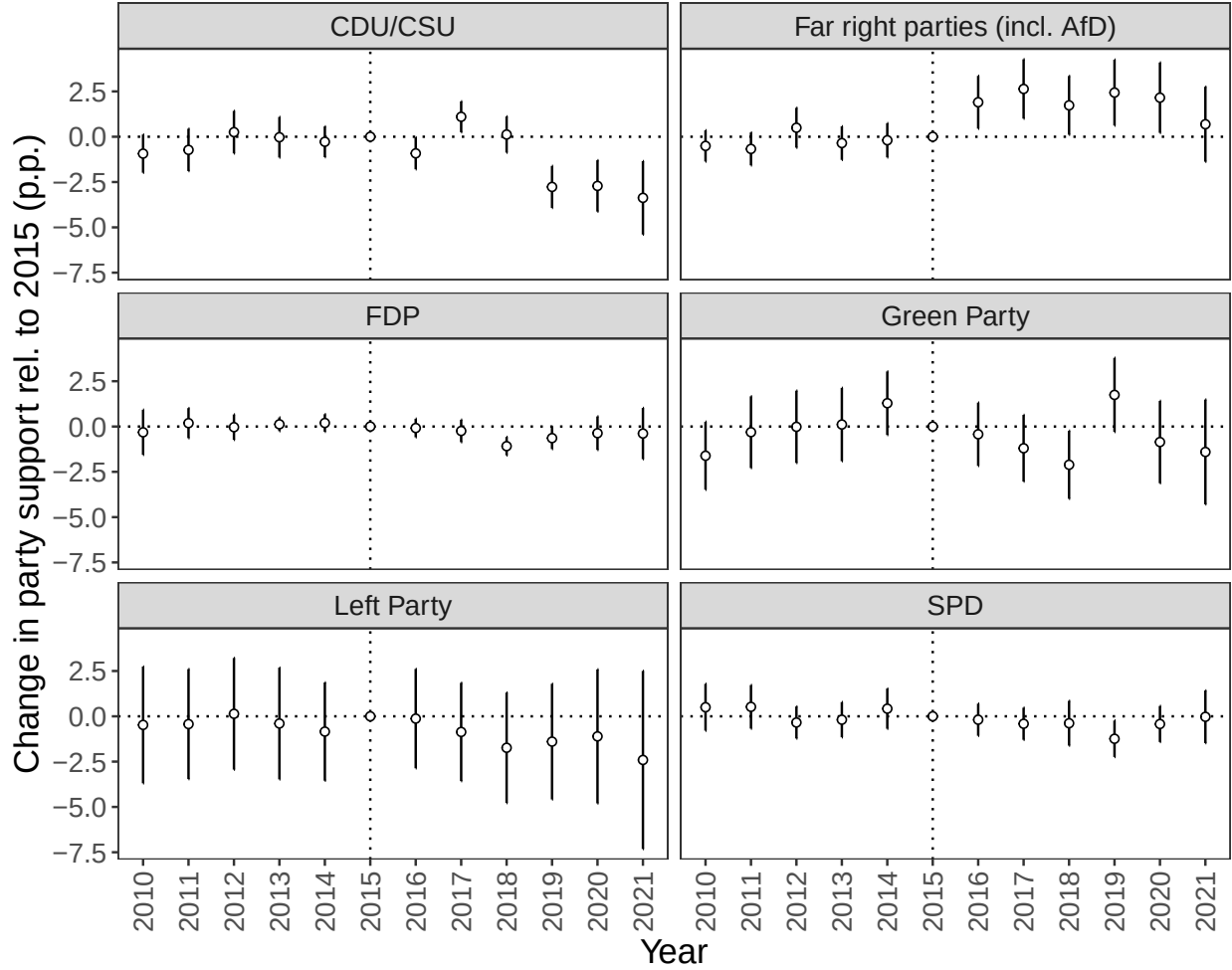
5.2 Individual-level results

In the previous section, we have established that counties with a higher share of workers in brown jobs experience larger increases in support for the AfD after 2013. We now use individual-level panel data to examine whether we see similar effects when comparing individuals in brown vs. non-brown occupations. In figure 7, we further present results from specification 3 in section 4.2. We find that support for far-right parties (mainly the AfD) increases after 2015. We observe losses for the CDU/CSU, as well as suggestive evidence for losses among the Green Party, particularly in the first few years after 2015. These results align with a descriptive assessment of partisan support among individuals in brown vs. non-brown occupations, which we show in figure A.4 in the SI. Further, the results shown in figure 7 are similar when we estimate an alternative specification that interacts brown employment status with an indicator for the post-2015 period, as we show in table A.6 in the SI.

As discussed previously, the fact that individuals can and do switch out of brown occupations means that the composition of the groups defined by brown and non-brown employment is not constant over time. The results presented in figure 7 as well as table A.6 address this through the inclusion of time-varying covariates and individual fixed effects. In table A.7 and table A.8, we further show that the main result is robust to the inclusion of weights that (partially) account for compositional differences (see also section A.3 for more details on the weights).

Heterogeneous treatment effects: We further assess whether the individual-level effects vary across dimensions that are often shown to be salient for vote choice: gender, age, income, and education. In table A.10 in the SI, we show that, for those in brown occupation, the shift towards the AfD is particularly pronounced among men and respondents older than 50 years. For income and education, the observed interaction effects do not permit clear conclusions.

Figure 7: Brown occupations and partisan support over time



Notes: The figure shows estimated coefficients based on the specification in equation 3. All models include base covariates. Standard errors are clustered at the individual level.

Comparing aggregate and individual effects: Because we present panel evidence at both the aggregate and individual level, we can directly examine whether individual-level vote shifts are sufficient to explain aggregate patterns. At the individual level, we find that workers in high-emission jobs are 2-3 percentage points more likely to support the AfD after 2015 (see table A.6). That is, among 100 employees in these jobs, the AfD attracts 2-3 additional voters compared to their counterparts in greener occupations. When examining county-level vote totals, we find that a one percentage point increase in the proportion of

brown jobs within a region in 2013 has an effect of approximately 0.2 to 0.3 additional percentage points in AfD between the 2013 and 2017 federal elections (see figure 4 and table A.3). If the aggregate effects are solely due to vote switching among employees in brown occupations, the aggregate effects suggest that 20-30 out of every 100 employees in brown jobs shifted their vote to the AfD.

The discrepancy between these figures suggests that the aggregate impact cannot be entirely attributed to shifts in voting behavior among those working in brown occupations. Therefore, one implication of our results is that not only individuals in brown jobs, but also those in non-brown occupations within areas more reliant on high-emission industries switch towards the far-right. Note that the observed differences in effects are not attributable to variations in treatment definition between aggregate and individual-level analysis because we use the same rule for coding high-emission jobs (see section 3.1).

6 Robustness

Finally, we conduct a number of robustness checks for both aggregate and individual-level analyses.

Aggregate-level results: First, we show that our results are robust to alternative definitions of the aggregate-level treatment in figure A.7 in the SI. Second, we show that the results are not driven by one specific state – excluding states one-by-one does not change our main conclusions (see figure A.8 in the SI). Third, we re-estimate the main specification using population weights, and show that our results remain largely similar to the results in the main specification (figure A.11). Fourth, we use the Sun and Abraham (2021) estimator to account for potential bias due to treatment effect heterogeneity in difference-in-differences settings. In figure A.6, we find that our conclusions are not affected by this additional robustness check. Fifth, we show that our results are not dependent on the inclusion of administrative district-specific election fixed effects or covariates. In table A.4, we find that results are sim-

ilar using either state-election fixed effects, or simply election fixed effects that do not vary by state or administrative district. We further show that the results remain similar when excluding the covariates listed in section 4.1. Sixth, we use an alternative treatment based on a variable constructed by Südekum and Rademacher (2024). This variable measures the county-level share of full-time employees in industries that experienced an increase in CO2 emissions between 2000 and 2019 (see section A.2.1 for more information). The correlation between this variable and our main measure of brown job employment is relatively high, at about 0.59. In figure A.10, we find very similar results when replacing our main treatment with the Südekum and Rademacher (2024) measure.

Individual-level results: In models 4 and 7 in table A.7, we assess whether regional differences can account for the observed individual-level relationship between brown occupations and radical-right support. Akin to county-level specification, we do so by including state-specific time trends, and find that our results remain largely unchanged. We further show that the results are unchanged when we do not use a time-varying treatment for working in a brown job, but instead use a treatment that is based on whether an individual had a brown job in 2015. In models 7–8 of table A.6, we find that our results are, if anything, slightly stronger when we fix the brown job indicator to its 2015 value.

7 Conclusion

The transition to sustainable energy sources is crucial to combating climate change. However, the shift from non-renewable to renewable energy sources is not only an environmental and economic shift, but also a profound social and political transformation. We argue that workers in traditional, non-renewable industries and occupations face economic and social difficulties, as transitioning to greener jobs is often challenging. We expect that workers in “brown” jobs will favor political actors that oppose energy transition policies.

We provide evidence for our argument from Germany, a country that has long pursued

extensive energy transition policies, and where the far-right AfD has been the sole vocal opponent of these policies. Using a difference-in-differences design and county-level and individual-level panel data, we show that workers in brown occupations disproportionately shifted towards the AfD party began vocally opposing energy transition policies in 2016.

Based on our findings, we propose several implications and potential avenues for future research. First, our study highlights the need for policymakers to consider the broader socio-political impacts of energy transition policies, particularly on workers in affected industries. This calls for comprehensive strategies that not only promote environmental sustainability, but also address the economic and social challenges faced by these workers. Recent scholarship shows that compensatory policies in the Spanish coal country can alleviate economic and political grievances (Bolet, Green and González-Eguino, 2023). Future research could test similar programs more broadly, focusing on other occupations and industries. This could include examining the role of retraining programs, economic incentives, and community engagement initiatives in mitigating the adverse effects of this transition.

Second, our research underscores the importance of understanding the dynamics of political allegiance shifts in the context of major policy changes. The pronounced movement toward the far-right observed among workers in 'brown' occupations suggests a deeper underlying narrative of perceived economic vulnerability and identity threat.

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A Supplementary Materials

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A.1 Additional information on the AfD platform change

We now identify and discuss some explanations of the AfD platform change that is described in the main body of the paper.

First, the platform change could have been the result of the change in leadership in the party and a concomitant shift towards the far-right. In July 2015, Frauke Petry replaced Bernd Lucke and shifted the party from its eurosceptic origins towards a more national-conservative and anti-immigration posture. The leadership change not only allowed for more right-wing views to be expressed within the party’s structure, but also heightened its tendency to oppose policies endorsed by the government and all mainstream parties (Küppers, 2024; Spoon, Hobolt and de Vries, 2014). An 2016 internal strategy paper of the AfD leadership outlined that the AfD’s opposition to the energy transition and environmental policies is primarily tactical, aiming to establish a distinct identity rather than being rooted in substantive disagreement. This approach aims to differentiate the AfD from parties like the Greens, whom it views as its main political adversary, using climate change as a key issue for setting itself apart from all other parties and the prevailing media narrative (AfD Bundesvorstand, 2016; Selk, Kemmerzell and Radtke, 2019, p. 55). This opposition is also part of a broader political narrative seeking to return to a “good old” industrial era, viewing the energy transition as a symbol of disliked post-industrial reforms and post-materialist values, against which right-wing populism is rallying.

Second, the AfD’s clearer opposition to climate change and energy transition policies could have resulted from a broader professionalization process within the party. The development of their Grundsatzprogramm, a result of extensive work by working groups, reflected a move toward more strategic, complex, and streamlined policy positions.¹⁹ This professionalization suggests that the party’s platform shifts are calculated rather than merely reactionary.

Third, the change in the AfD’s stance coincided with a global trend among right-wing parties and actors who had adopted climate-skeptic positions. The success of figures like Donald Trump, who ran on a climate-skeptic platform in 2015 and 2016, alongside similar movements worldwide, likely served as a motivation for the AfD.

Finally, the change in the party’s stance might be a direct response to voter grievances and demand for policy change. However, our analysis below indicates that a trend towards the AfD was not apparent among workers in high-emission jobs that are vulnerable to the energy transition (see section 5.1). Hence, we conjecture that the AfD’s policy change was likely the result of a combination of the three previous elements, in particular the strategic opposition to the policy stance of all other major parties.

A.2 Data

A.2.1 Alternative treatment

Our main treatment is based on emissions and occupation data from the US context. To address concerns about the applicability of this measure to the German context, we probe

¹⁹<https://www.dw.com/de/was-die-afd-wirklich-will-parteitag-soll-entscheiden/a-19217147>

the robustness of our results to an alternative measure based on German emissions and employment data.

We use a measure of county employment in “problem sectors” developed by Südekum and Rademacher (2024). The measure is based on national CO2 emissions data derived from the Federal Statistical Office of Germany and county-level employment data from IAB (Institut für Arbeitsmarkt- und Berufsforschung). In particular, for each industry (Wirtschaftszweig), nationally aggregated total CO2 emissions are measured for the period 2000 to 2019. Industry emissions are then combined with employment data for each of these industries in a given county and year.

We use a variable provided by Südekum and Rademacher (2024) which measures the share of full-time employees in industries that experienced an increase of CO2 emissions between 2000 and 2019 measured for each county in 2019:

$$D_c = \frac{\sum_{a=1}^A N_{ac} \times \mathbf{1}[(\text{CO2}_{a,2019} - \text{CO2}_{a,2000}) > 0]}{\sum_{a=1}^A N_{ac}}$$

A.2.2 Election data

County mergers: During the period of time we study, the county borders change. To create a balanced panel where the number of counties remains constant over time, we map the boundary of each county in the years before 2021 to its 2021 boundaries using crosswalk files from the Federal Institute for Research on Building, Urban Affairs, and Spatial Development (BBSR). Specifically, for all counties that were subject to change, we calculate a weighted average of each variable that we use in the analyses using a population-weighted key. For example, if counties a and b merge into a new county c , the value of a variable X_c^{merged} for county c after merging is the population-weighted average of the values X_a and X_b , that is, $X_c^{merged} = \frac{w_a}{w_a + w_b} X_a + \frac{w_b}{w_a + w_b} X_b$, where w_a and w_b are the population sizes of counties a and b . The units of analysis are counties in their 2021 boundaries. This procedure may introduce some measurement error in our models.

A.2.3 Anti-refugee violence and demonstrations data

To assess whether our main results reflect the activation of xenophobia in regions with counties with a higher share of high-emission jobs, we obtain georeferenced data on categorized events of anti-refugee violence and social unrest in Germany from 2014 to 2018 provided by Benček and Strasheim (2016). The set of event data is based on a public chronicle maintained by the civil society project “Mut gegen rechte Gewalt” (“Courage against right-wing violence”). This dataset encompasses 6,918 distinct events identified as politically motivated attacks against refugees over four years. We measure the number of anti-refugee events for each county in 2015 because 2015, the last year before the AfD started campaigning on opposition to energy transition policies.

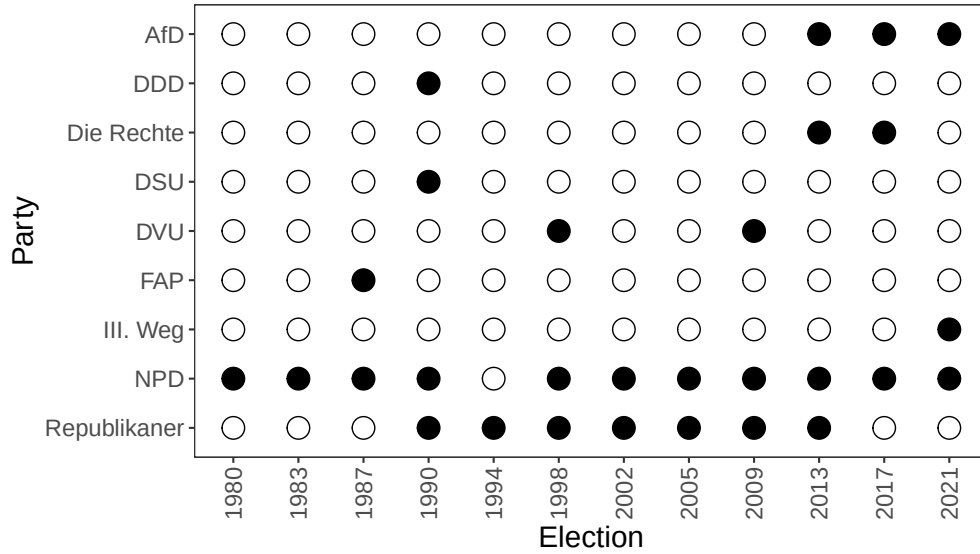
To account for the fact that anti-refugee violence might be driven by the size of the pool of potential perpetrators or by the overall crime propensity in a specific county, we scale this measure by a county’s total population and the total crime count in 2015. Specifically, we generated two metrics: the number of anti-refugee incidents per 10,000 inhabitants, and the

number of anti-refugee incidents per 1,000 total crimes reported in each county.

A.2.4 Far-right parties

Our main analyses rely on an outcome measure that is the sum of several far-right parties. This outcome comes with the advantage that we can observe trends in voting prior to 2013 in federal elections, which is not possible for the AfD itself. We follow the coding of the party family by the ParlGov project (Döring, Huber and Manow, 2023). Figure A.1 shows which far-right parties competed in German federal elections between 1980 and 2021.

Figure A.1: Far-right parties competing in German federal elections



Notes: Solid dots indicate that a given far-right party competed in a particular German federal election. Party abbreviations: AfD = Alternative für Deutschland, DDD = Bund der Deutschen Demokraten, DSU = Deutsche Soziale Union, DVU = Deutsche Volkunion, FAP = Freiheitliche Deutsche Arbeiterpartei, NPD = Nationaldemokratische Partei Deutschlands.

A.3 Weighting

Compositional differences: As described in section 4, the composition of the group of respondents in brown occupations changes over time. In particular, the share of respondents in brown occupations decreases over time. Formally, for some covariate X , we can write the mean of this variable for respondents in brown and non-brown occupations in year t as:

$$\mu_{X,\text{brown},t} = \frac{\sum_{i=1}^N X_{it} D_{it}}{\sum_{i=1}^N D_{it}}$$

$$\mu_{X,\text{non-brown},t} = \frac{\sum_{i=1}^N X_{it} (1 - D_{it})}{\sum_{i=1}^N (1 - D_{it})}$$

We can call the difference between these two means $\Delta\mu_{X,t} = \mu_{X,\text{brown},t} - \mu_{X,\text{non-brown},t}$. As stated above, $\Delta\mu_{X,t}$ is not constant across t . In designs where D_{it} is fixed across all t , the variance of $\Delta\mu_{X,t}$ across t can only stem from changes in X_{it} , i.e. from time-varying covariates – in our design, variance in $\Delta\mu_{X,t}$ can also stem from changes in the composition of the group defined by D_{it} .

Calculating weights: To calculate weights, we proceed as follows:

- We select 2011 as the reference year. We then select all observations that are either observed in 2011 or in a year t' such that $t \in \{2011, t'\}$ with $t' \neq 2011$. We further restrict the set of observations to $D_{it} = 1$, which means we end up with the set $\{(i, t) \mid t \in \{2011, t'\} \text{ and } D_{i,t} = 1\}$.²⁰
- We then define a simple indicator variable T_{it} that tells us whether $t = 2011$ or $t = t'$, i.e. $T_{it} = \mathbb{1}\{t = 2011\}$. Next, we estimate the propensity of an individual i being observed in 2011 as a function of all covariates. We call this propensity score $\hat{\pi}_{it'}(X_{it})$, which measures the probability of $T_{it} = 1$.
- For observations with $t = t'$, we then obtain weights $w_{it'} = 1/(1 - \hat{\pi}_{it})$.
- We repeat the previous steps for the set of observations that are in non-brown jobs, i.e. for the set $\{(i, t) \mid t \in \{2011, t'\} \text{ and } D_{it} = 0\}$. This gives us weights $w_{it'}$ such that $t' \neq t$. For individual observed in $t = 2011$, we set $w_{it} = 1$ for all i .

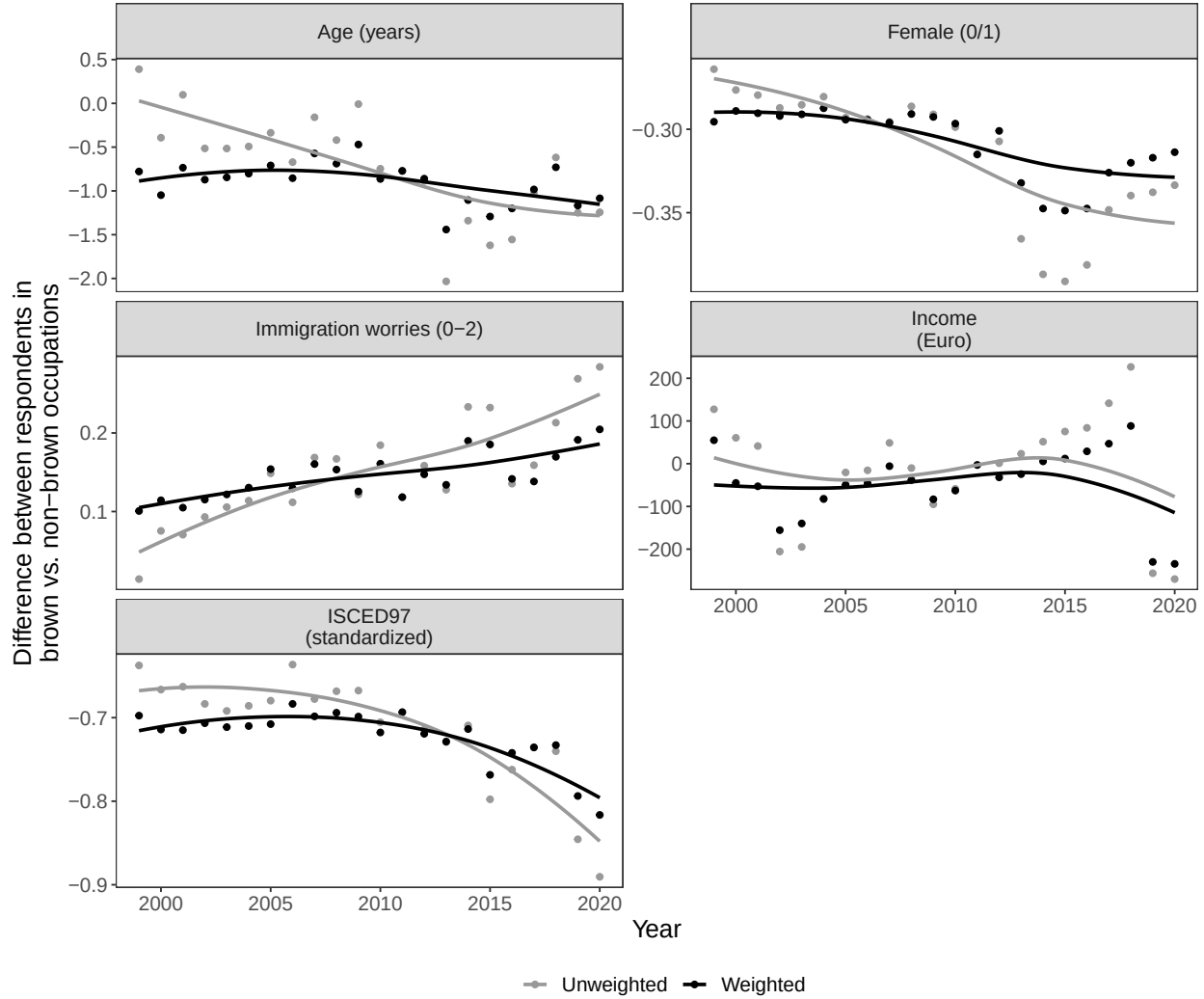
Intuitively, this approach selects a target sample (either individuals in brown jobs or individuals in non-brown jobs in 2011), and then weights individuals observed in all other years to reduce the change in the composition of the brown and non-brown groups over time.

Compositional differences in the weighted sample: Given the procedure described above, we obtain weights w_{it} . We can then calculate weighted versions of $\mu_{X,\text{brown},t}$ and $\mu_{X,\text{non-brown},t}$, and then obtain the difference in weighted means $\Delta\mu_{X,t}^{\text{weighted}}$.

In figure A.2, we present both $\Delta\mu_{X,t}$ and $\Delta\mu_{X,t}^{\text{weighted}}$. The relevant metric is whether the differences between brown and non-brown groups are changing over time. Put differently, if the slope of the relationship between year and the mean covariate difference between

²⁰For clarity, we can also write this set as $\{(i, t) \mid t = 2011 \text{ and } D_{i,2011} = 1\} \cup \{(i, t) \mid t = t' \text{ and } D_{i,t'} = 1\}$

Figure A.2: Differences between respondents in brown and non-brown occupations over time



Note: The figure shows the average difference between individuals in brown and non-brown occupations over time, for five covariates. We present weighted and unweighted differences. For more information, see A.3.

groups $D_{it} = 1$ and $D_{it} = 0$ is different from 0, this indicates that the composition of brown occupations changes relative to non-brown occupations. We find evidence for changes in composition for all covariates. After weighting, these changes are noticeably less pronounced, but not absent.

A.4 Descriptives

A.4.1 High-emission jobs

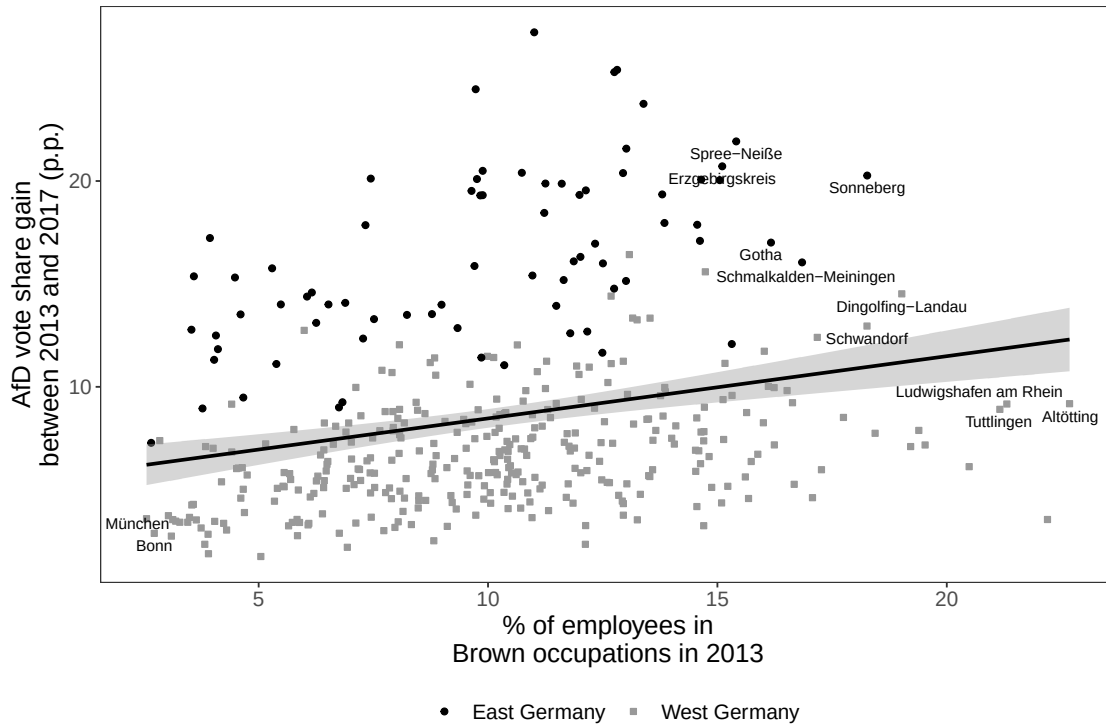
Table A.1: Most frequent high-emissions jobs in the 2015 SOEP

ISCO08 Code	ISCO08 Title	N
7233	Agricultural and industrial machinery mechanics and repairers	184
7223	Metal working machine tool setters and operators	116
7222	Toolmakers and related workers	65
8160	Food and related products machine operators	65
8131	Chemical products plant and machine operators	50
7112	Bricklayers and related workers	49
8189	Stationary plant and machine operators not elsewhere classified	46
3111	Chemical and physical science technicians	44
8342	Earthmoving and related plant operators	33
8121	Metal processing plant operators	30

Notes: The table shows the most frequent high-emissions jobs in the 2015 SOEP.

A.4.2 Aggregate-level data

Figure A.3: Change in AfD support 2013–2017 and brown occupations in 2013



Notes: The figure shows the county-level shares of brown occupations in 2013 and the change in AfD support between 2013 and 2017. We highlight select counties.

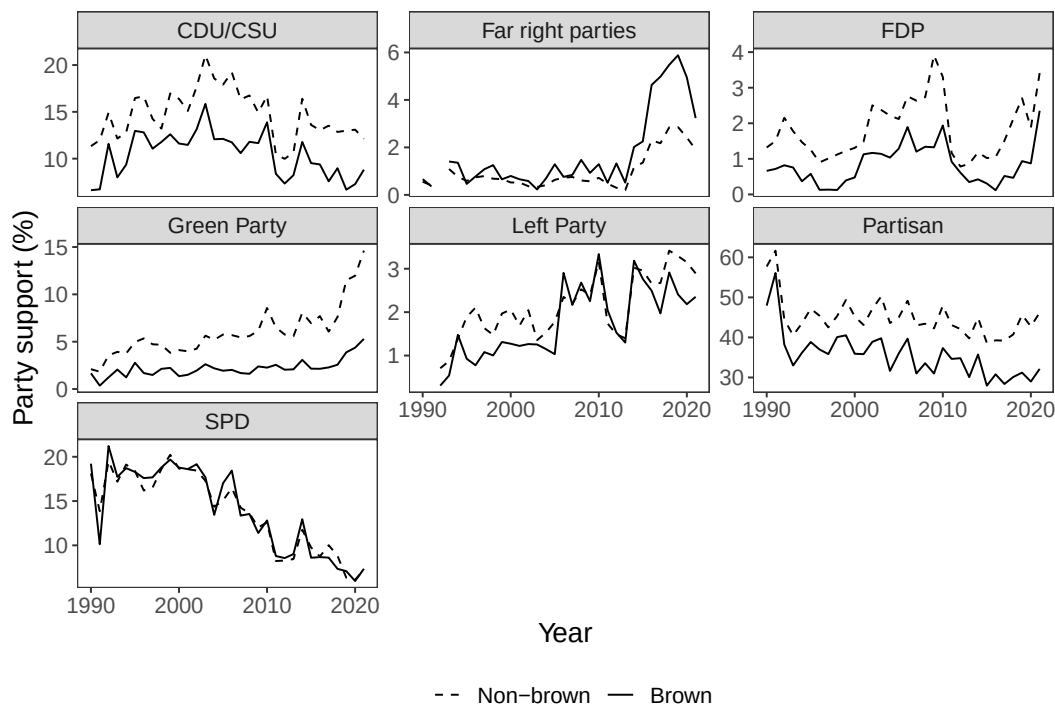
A.4.3 Individual-level data

Table A.2: Descriptive statistics, comparing respondents in brown jobs with respondents in non-brown jobs in 2015

Brown job	Variable	Mean	Observations
No	Age	43.12	14,577
Yes	Age	41.49	977
No	Female	0.53	14,577
Yes	Female	0.15	977
No	ISCED97	4.01	14,257
Yes	ISCED97	3.21	954
No	Manufacturing job	0.18	14,577
Yes	Manufacturing job	0.71	977
No	Monthly income	2619.42	13,333
Yes	Monthly income	2686.00	880

Notes: The table shows average for a set of variables, separately for brown- and non-brown occupations. The sample is SOEP respondent in regular employment in 2015. Differences in sample sizes are due to missing values.

Figure A.4: Brown occupations and partisan support, before and after 2015



Notes: The figure present average party support among SOEP respondents in brown and non-brown occupations over time. For more information on the definition of a brown occupation, see section 3.1.

A.5 Additional county-level results

Table A.3: Main effects

	SPD (1)	FDP (2)	Greens (3)	Left party (4)	AfD (5)	CDU/CSU (6)	Far-right (7)
Brown share $\times$ Election = 2017	0.071 (0.037)	-0.056** (0.020)	-0.030 (0.018)	-0.095*** (0.025)	0.228*** (0.054)	-0.099 (0.051)	0.203*** (0.046)
Brown share $\times$ Election = 2021	0.046 (0.051)	0.034 (0.023)	-0.284*** (0.040)	-0.063** (0.022)	0.284*** (0.051)	-0.094 (0.068)	0.258*** (0.043)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election-Admin. district FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.989	0.979	0.976	0.990	0.978	0.987	0.981
Observations	2,313	2,313	2,313	2,313	1,194	2,313	2,313

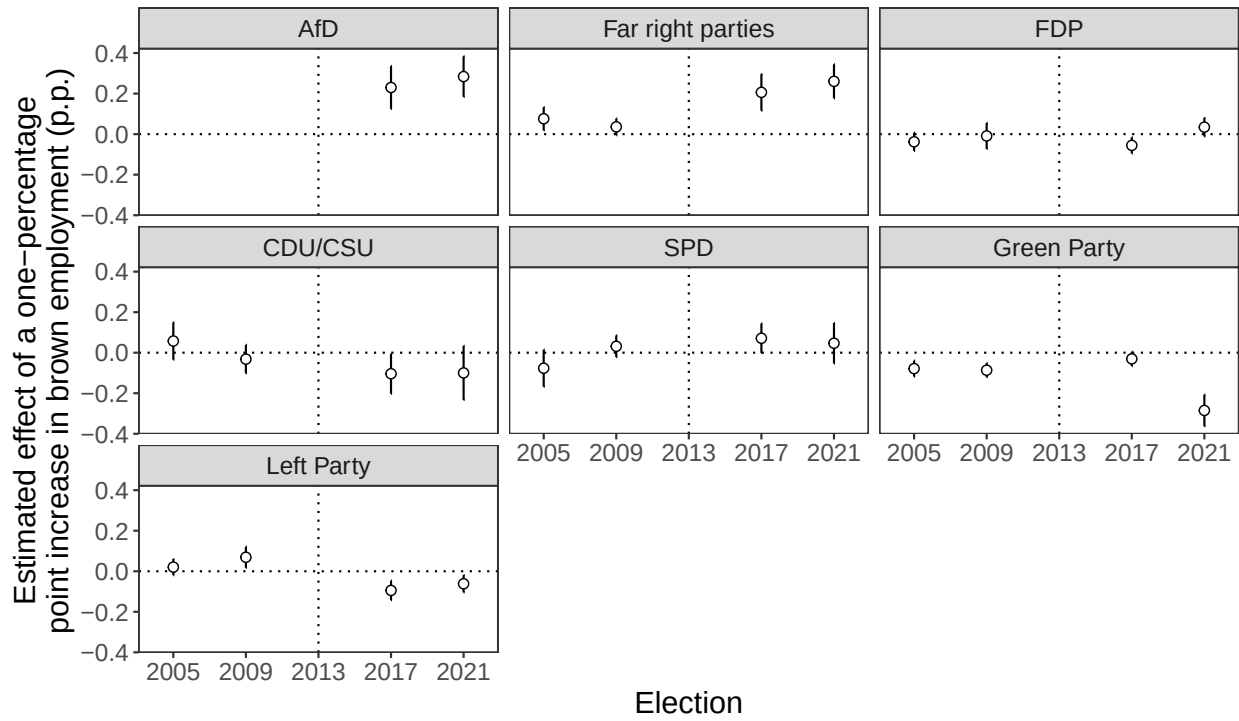
Notes: The table presents the results from the main specification. We present estimates for θ_0 and θ_1 from equation 2. The quantities listed here are the same ones as presented in figure 4, as well as figure 7. Standard errors are clustered by county. See section 4.1 for more information on the estimation. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

Table A.4: Additional specifications using alternative fixed effects

	DV: Far-right vote share			
	(1)	(2)	(3)	(4)
Brown share $\times$ Election = 2017	0.266*** (0.053)	0.211** (0.069)	0.179*** (0.052)	0.207*** (0.046)
Brown share $\times$ Election = 2021	0.383*** (0.059)	0.316*** (0.070)	0.244*** (0.043)	0.260*** (0.043)
Covariates	No	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Election FE	Yes	Yes	No	No
Election-State FE	No	No	Yes	No
Election-Admin. district FE	No	No	No	Yes
R ²	0.849	0.928	0.975	0.982
Observations	2,791	2,307	2,307	2,307

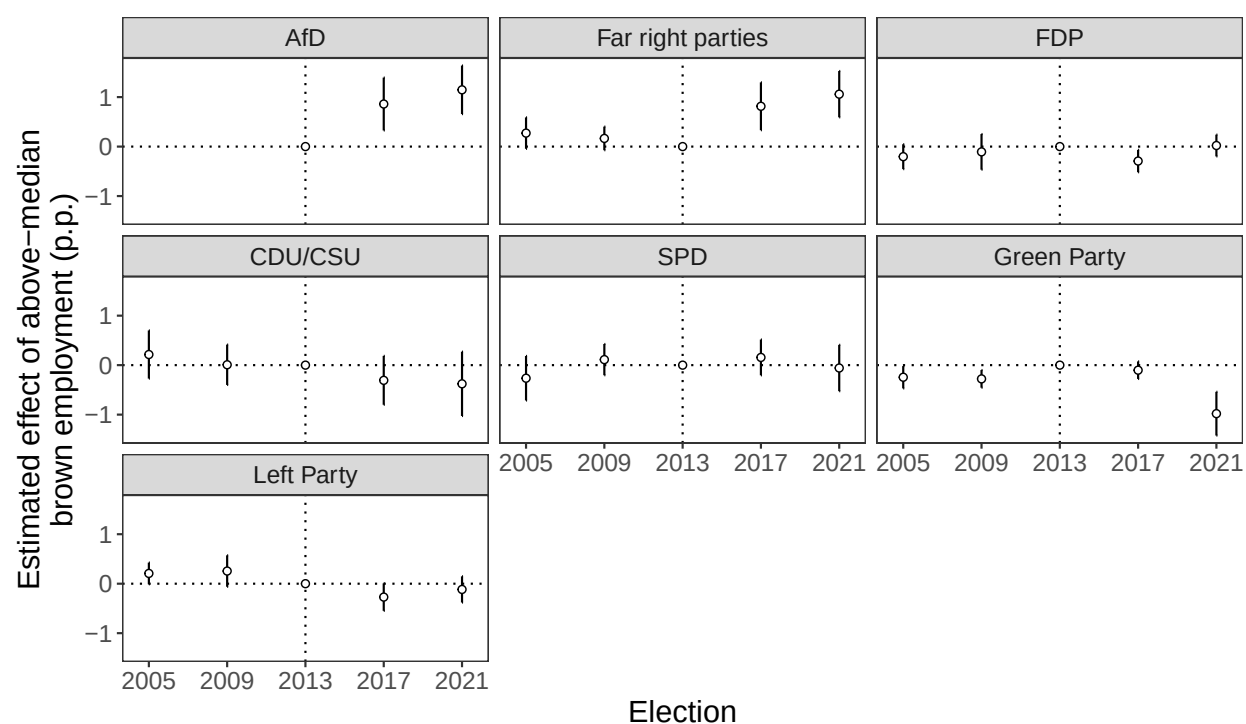
Notes: The table presents the results from the main specification. In contrast to table A.6, this table presents specification that uses different fixed effects. We present estimates for θ_0 and θ_1 from equation 2. Standard errors are clustered by county. See section 4.1 for more information on the estimation. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

Figure A.5: Estimated coefficients before and after 2013



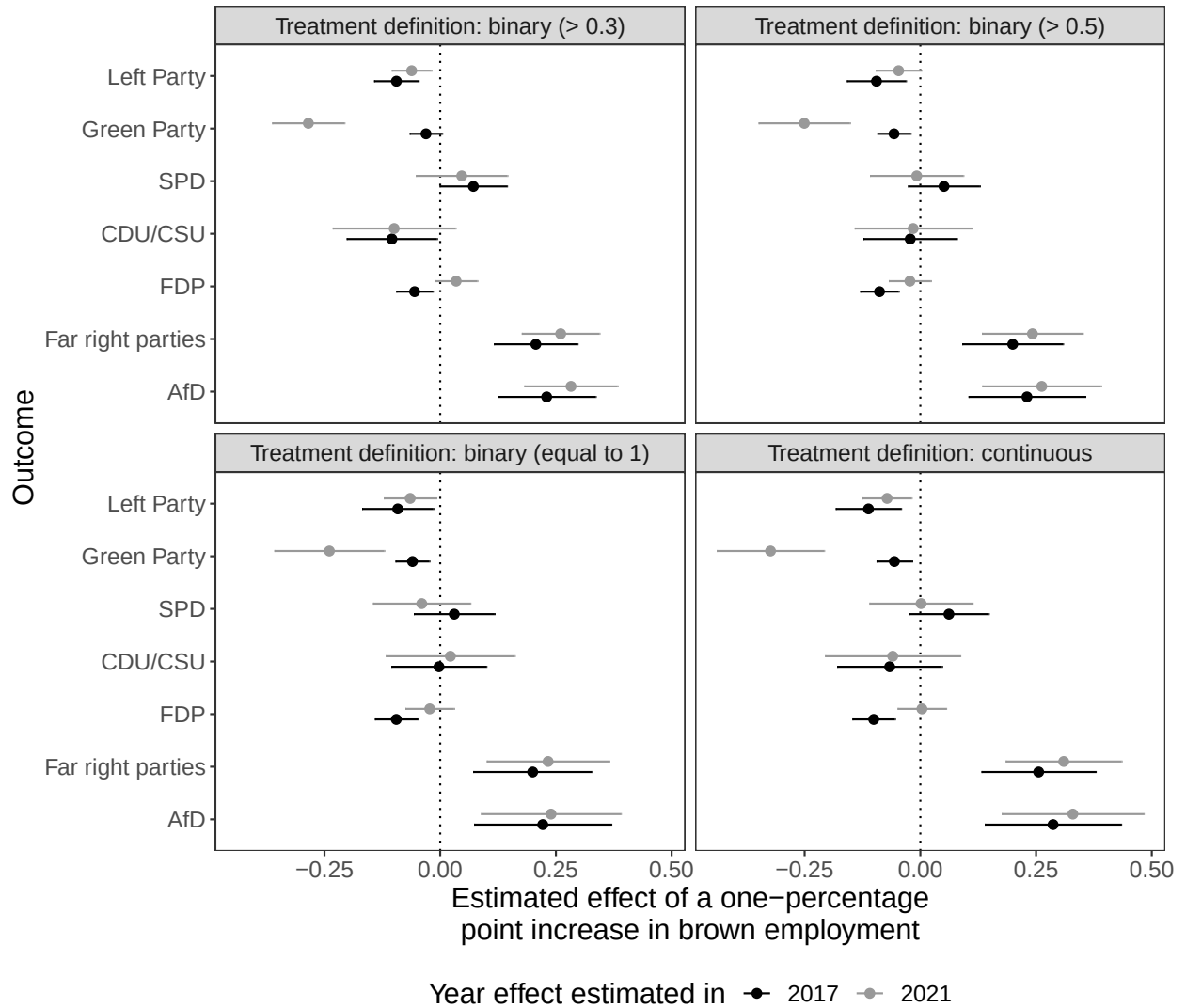
Notes: The figure presents the results from the main specification. As opposed to figure 4, where we only show θ_0 and θ_1 , this figure also contains the pre-treatment estimates θ_{-2} and θ_{-3} . Standard errors are clustered by county. See section 4.1 for more information on the estimation.

Figure A.6: Estimated coefficients before and after 2013, using the estimator by Sun and Abraham (2021)



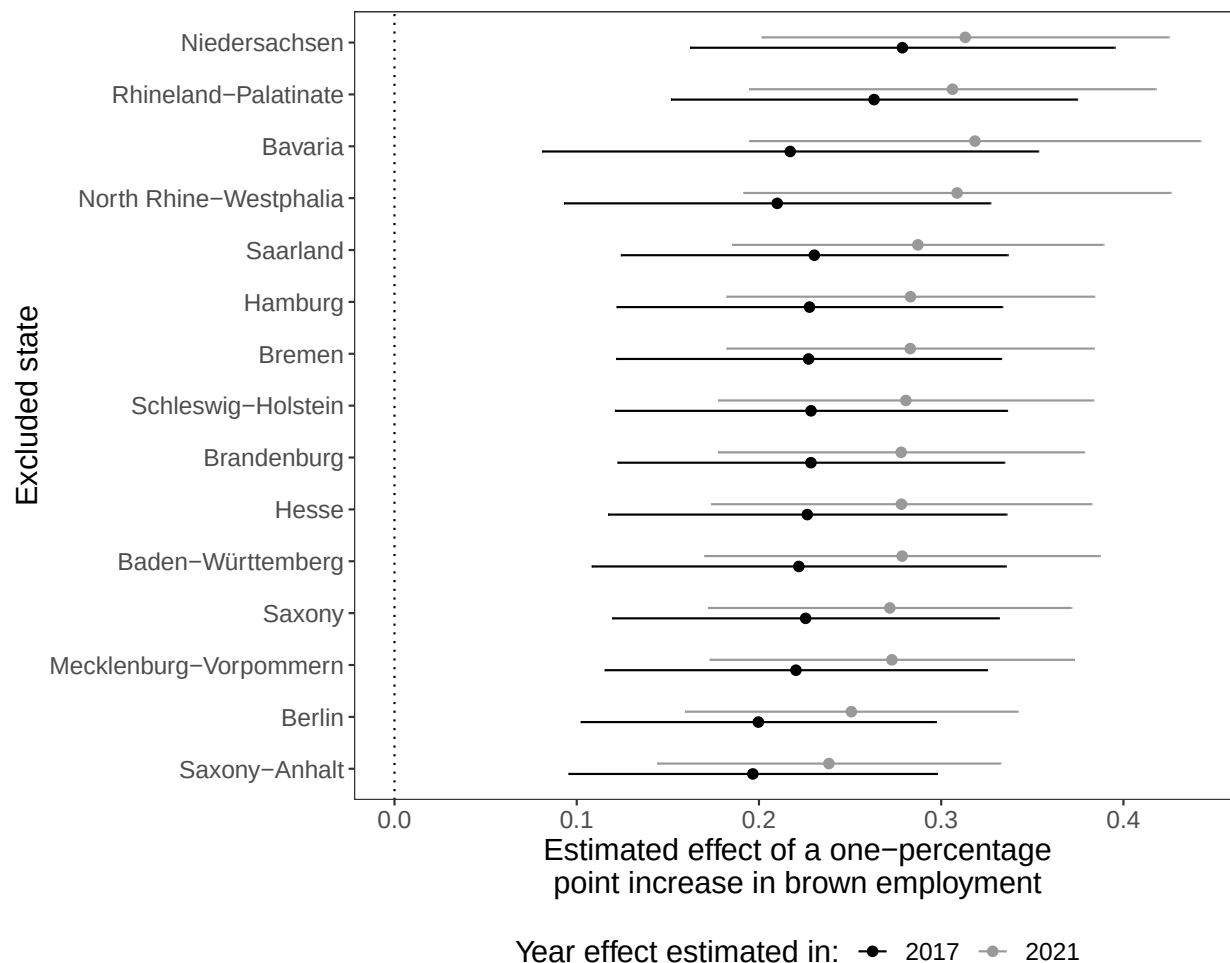
Notes: The figure presents the results from the main specification. Estimates are based on the Sun and Abraham (2021) estimator. The treatment is a binary variable for whether the share of people in brown employment is above the median share of brown employees in 2013 (the median is 10.1%).

Figure A.7: Aggregate results, for different treatment definitions



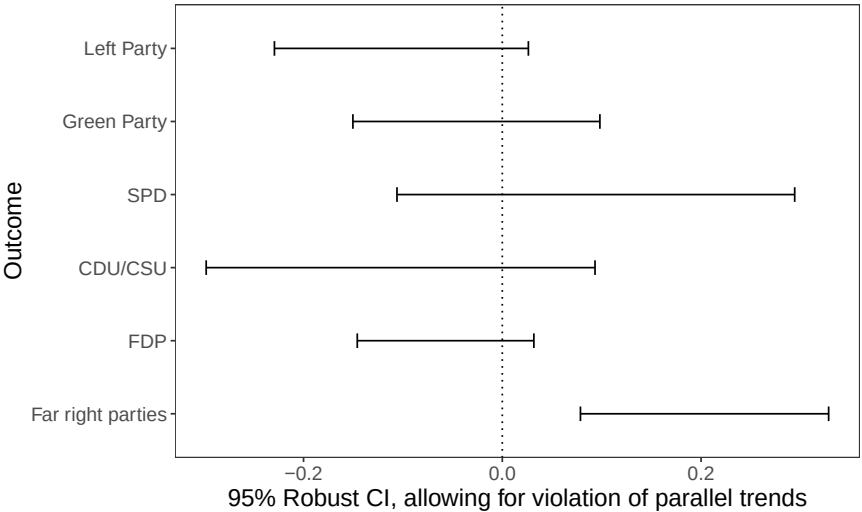
Notes: The figure presents the results from the main specification, for different treatment definitions. Co-variates and additional information on the estimation are listed in section 4.1. Standard errors are clustered at the county level.

Figure A.8: Aggregate results, excluding states one-by-one



Notes: The figure presents the results from the main specification, excluding states one-by-one. We only present results for the AfD. Covariates and additional information on the estimation are listed in section 4.1. Standard errors are clustered at the county level.

Figure A.9: Assessing parallel trends violations, based on Rambachan and Roth (2023)



Notes: the figure shows the estimated 95% confidence interval for the effect on county-level voting behavior, allowing for a parallel trends violations as severe as the ones observed in figure A.5. For details on this approach, see Rambachan and Roth (2023).

A.5.1 Changes in income, economic output, and unemployment

Below, we assess whether counties with higher shares of brown employment in 2013 saw differential trends in (i) household income, (ii) GDP/capita and (iii) unemployment rates. To do so, we estimate the specification outlined in equation 2, but replace the outcome with the three aforementioned variables. We present results from two specifications: the first includes only county fixed effects as well as administrative district—year fixed effects (i.e. only γ_c and δ_{jt}). The second further includes the $M_c \times \mathbf{1}_t = k$ and $P_c \times \mathbf{1}_t = k$ terms, as a well as the county population. The second specification aims at identifying economic changes in counties with many brown employees that are unrelated to manufacturing employment or urban-rural differences.

We present the results in table A.5. We find no evidence for decreases or increases in household income. We do find some evidence that GDP/capita decreases (in relative terms), as well as some evidence for increases in unemployment. However, the evidence on unemployment rates is not robust to account for urban-rural differences, as well as differences in manufacturing employment.

Regarding effect magnitude, going from the 25th to the 75th percentile of brown employment shares in 2013 (i.e. going from 7.2% to 12.6%) is associated with (relative) declines of at most 2,930 Euros in GDP/capita in 2021, which is equal to about nine percent of overall GDP/capita in 2013.

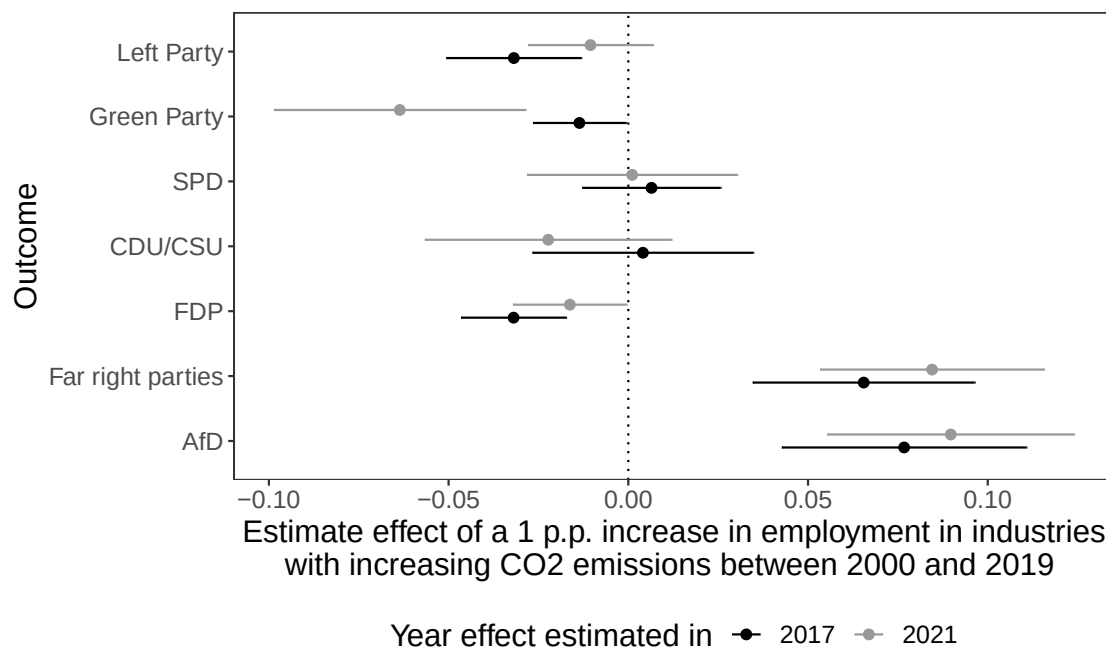
Table A.5: Changes in income and unemployment rates after 2013

	DV: HH income	DV: GDP/pc	DV: Unem- ployment rate	DV: HH income	DV: GDP/pc	DV: Unem- ployment rate
	(1)	(2)	(3)	(4)	(5)	(6)
Brown share $\times$ Election = 2017	19.1 (11.8)	17.3 (41.0)	0.018* (0.008)	17.3 (19.9)	-336.1* (132.1)	-0.011 (0.010)
Brown share $\times$ Election = 2021	19.8 (12.8)	-177.1** (57.7)	0.022* (0.010)	-33.2 (21.4)	-540.6** (205.2)	-0.010 (0.014)
DV mean in 2013:	20,159.5	32,633.7	6.43	20,159.5	32,633.7	6.43
Covariates	No	No	No	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Election-Admin. district FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.980	0.989	0.987	0.981	0.991	0.988
Observations	1,193	1,196	1,196	1,191	1,191	1,191
Within Adjusted R ²	0.003	0.028	0.010	0.042	0.177	0.046

Notes: The table presents the results from the main specification. We present estimates for θ_0 and θ_1 from equation 2. The outcomes are listed in first row. Income variables are measured in Euros. Specifications 4–6 include additional controls, which are (i) the interaction between 2013 population density and election dummies, (ii) the interaction between 2013 manufacturing employment shares and election dummies and (iii) a time-varying indicator of county population. Standard errors are clustered by county. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

A.5.2 Alternative treatment definition based on Südekum and Rademacher (2024)

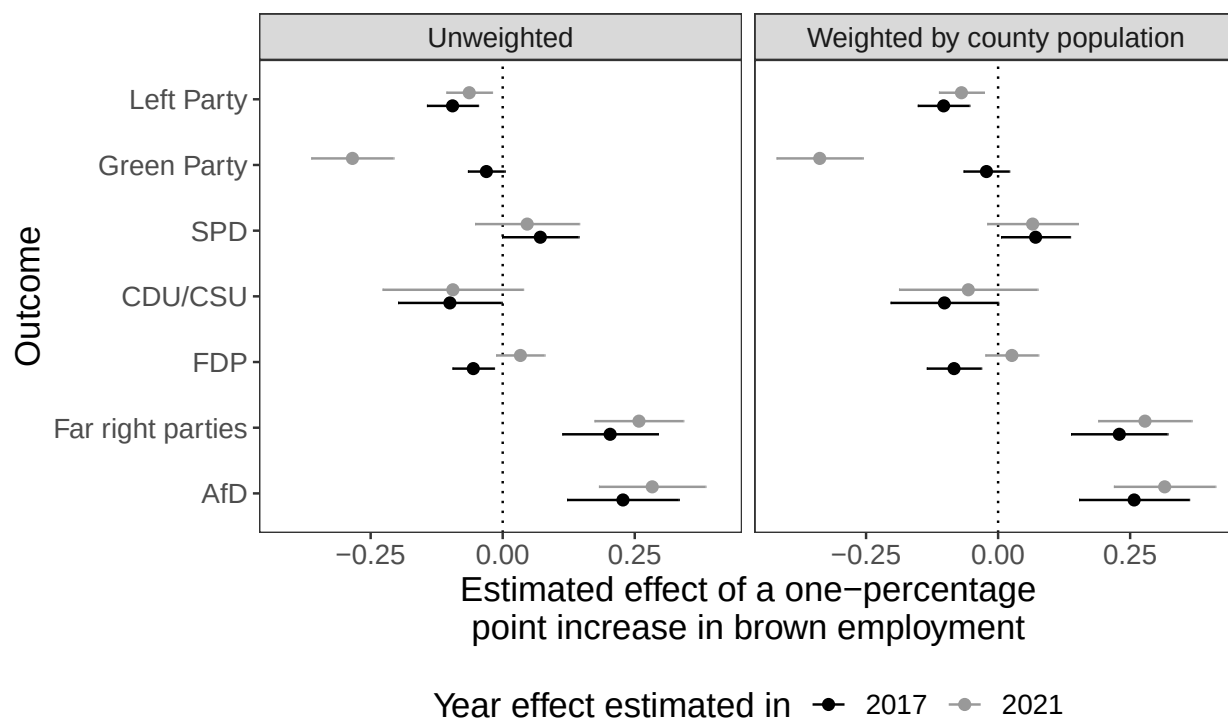
Figure A.10: Aggregate results



Notes: The figure presents the results from the main specification. Instead of the share of employees in brown occupations, we use employment shares in industries with increasing CO2 emissions between 2000 and 2019, obtained from (Südekum and Rademacher, 2024). See also section A.2.1 for more information on this variable. We present the coefficients θ_0 (for 2017) and θ_1 (for 2021) from the specification (equation 2) in section 4.1. Covariates are listed in section 4.1. Standard errors are clustered at the county level. The “far-right parties” term is the sum of all far-right parties, which includes that AfD. For details on the results, see table A.3.

A.5.3 Population weights

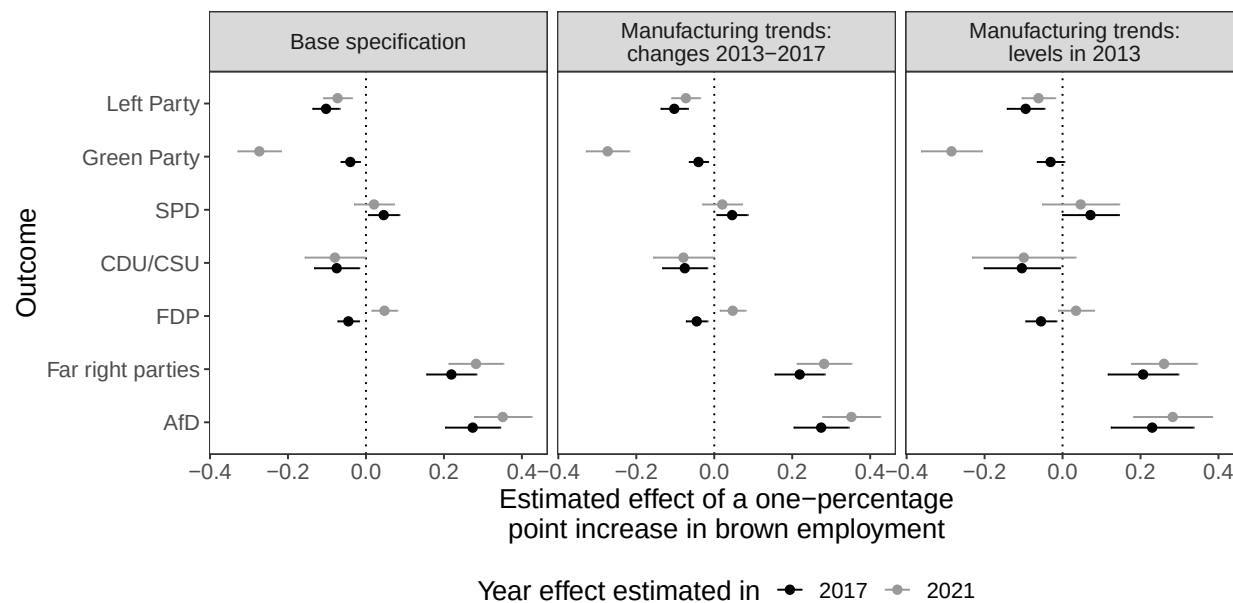
Figure A.11: Aggregate results, weighting counties by population



Notes: The figure presents the results from the main specification, for the base specification (left-hand side) and the specification that uses population weights (right-hand side). Covariates and additional information on the estimation are listed in section 4.1. Standard errors are clustered at the county level.

A.5.4 Manufacturing employment

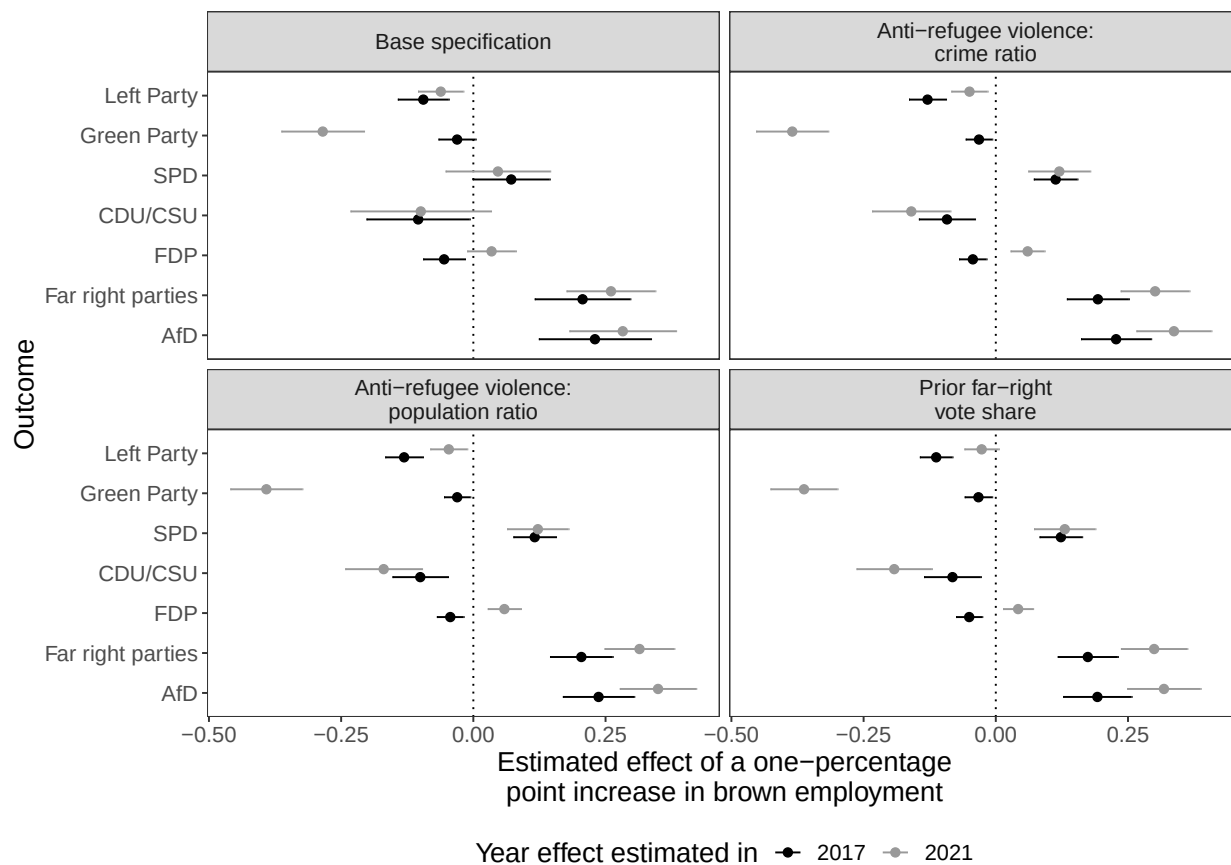
Figure A.12: Aggregate results, allowing for time-varying effects of manufacturing employment



Notes: The figure presents the results from the main specification, with an additional term that interacts pre-treatment share of employees in manufacturing with election year dummies. Covariates and additional information on the estimation are listed in section 4.1. Standard errors are clustered at the county level.

A.5.5 Existing anti-immigrant sentiments

Figure A.13: Aggregate results, allowing for time-varying effects of pre-2016 levels of anti-refugee violence



Notes: The figure presents the results from the main specification, with additional terms that interact election indicators with (i) the share of anti-refugee crimes among all crimes in 2015, (ii) the per-capita number of anti-refugee crimes in 2015 and (iii) the vote share of radical-right parties in the 2013 federal election. Covariates and additional information on the estimation are listed in section 4.1. Standard errors are clustered at the county level.

A.6 Additional individual-level results

The tables below present results for the individual-level specifications. Instead of the event study specification in the main body of the paper, we present results from a simpler specification that interacts the brown job indicator with an indicator variable for 2016 or later. The specification is as follows:

$$y_{it} = \tau D_{it} + \theta(D_{it} \times I_{2016+}) + \beta' X_{it} + \gamma_i + \delta_t + \varepsilon_{it} \quad (4)$$

This specification is similar to the one shown in section 4.2. The tables below contain estimates for the coefficient θ .

Table A.6: Individual-level specifications

	DV: SPD	DV: FDP	DV: Gruene	DV: Linke	DV: CDU/CSU	DV: Far-right support	DV: Supports any party
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Brown job $\times$ 2016+	-0.002 (0.007)	-0.0010 (0.002)	-0.005 (0.004)	-0.007 (0.004)	-0.005 (0.007)	0.018*** (0.006)	-0.0006 (0.011)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.682	0.603	0.718	0.664	0.713	0.528	0.657
Observations	178,993	178,993	178,993	178,993	178,993	178,993	178,285

Notes: The table presents results from the specification described above. The sample is all responses between 2010 and 2021. Standard errors are clustered at the individual level. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

Table A.7: Individual-level specifications – weights and constant job treatment

	DV: Far-right support (0/1)						
	Brown job: time-varying				Brown job: constant		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Brown job $\times$ 2016+	0.019*** (0.005)	0.018*** (0.006)	0.020** (0.007)	0.018** (0.006)	0.026*** (0.006)	0.028*** (0.007)	0.027*** (0.007)
Weights	No	No	Yes	No	No	No	No
Covariates	No	Yes	Yes	Yes	No	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No	Yes	Yes	No
Year-State FE	No	No	No	Yes	No	No	Yes
R ²	0.484	0.528	0.994	0.530	0.370	0.441	0.444
Observations	226,521	178,993	176,538	178,993	157,639	115,837	115,837

Notes: The table presents results from the specification described above. The sample is all responses between 2010 and 2021. Models 5–7 use a constant treatment based on whether the responded held a brown job in 2015. Standard errors are clustered at the individual level. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

Table A.8: Individual-level specifications – respondents observed in 2015

DV: Far-right support (0/1)							
	Brown job: time-varying				Brown job: constant		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Brown job $\times$ 2016+	0.020*** (0.005)	0.018** (0.006)	0.019** (0.007)	0.017** (0.006)	0.026*** (0.006)	0.028*** (0.007)	0.027*** (0.007)
Weights	No	No	Yes	No	No	No	No
Covariates	No	Yes	Yes	Yes	No	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No	Yes	Yes	No
Year-State FE	No	No	No	Yes	No	No	Yes
R ²	0.393	0.442	0.991	0.446	0.370	0.441	0.444
Observations	137,115	113,599	112,497	113,599	157,639	115,837	115,837

Notes: The table presents results from the specification described above. The sample is all responses between 2010 and 2021. Models 5–7 use a constant treatment based on whether the respondent held a brown job in 2015. All specifications include only respondents for which brown job status is observed in 2015. Standard errors are clustered at the individual level. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

Table A.9: Effects on individual-level attitudes

	Satisfaction: work (1)	Worried: job security (2)	Worried: eco- nomic devel- opment (3)	Worried: fi- nances (4)	Satisfaction: household income (5)	Satisfaction: individual income (6)
Brown job $\times$ 2016+	-0.025 (0.024)	-0.046 (0.026)	0.042 (0.024)	0.005 (0.022)	-0.022 (0.019)	-0.017 (0.018)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.545	0.604	0.554	0.620	0.678	0.669
Observations	173,195	174,954	178,774	178,872	176,140	176,909

Notes: The table presents results from the specification described above. All outcomes are standardized. Standard errors are clustered at the individual level. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

A.6.1 Heterogeneity

Table A.10: Additional individual-level specifications – heterogeneity

	DV: Far-right support (0/1)			
	(1)	(2)	(3)	(4)
Brown job $\times$ 2016+	0.018* (0.007)	0.006 (0.005)	0.018* (0.007)	0.018 (0.010)
Brown job $\times$ Female	0.009 (0.006)			
2016+ $\times$ Female	-0.014*** (0.002)			
Brown job $\times$ 2016+ $\times$ Female	-0.020* (0.010)			
Brown job $\times$ Above 50		-0.008 (0.008)		
2016+ $\times$ Above 50		0.0007 (0.002)		
Brown job $\times$ 2016+ $\times$ Above 50		0.038** (0.015)		
Brown job $\times$ Abitur			0.007 (0.006)	
2016+ $\times$ Abitur			-0.010*** (0.002)	
Brown job $\times$ 2016+ $\times$ Abitur			-0.012 (0.015)	
Brown job $\times$ High income				0.006 (0.008)
2016+ $\times$ High income				-0.002 (0.002)
Brown job $\times$ 2016+ $\times$ High income				0.0001 (0.013)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
R ²	0.474	0.474	0.457	0.427
Observations	140,401	139,663	111,927	114,331

Notes: The tables presents results from interaction specifications. Standard errors are clustered at the individual level. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

Table A.11: Additional individual-level specifications – heterogeneity by prior attitudes

	DV: Far-right support (0/1)		
	(1)	(2)	(3)
Brown job $\times$ 2016+	0.019** (0.006)	0.018** (0.006)	0.014* (0.006)
Brown job $\times$ Worried: job security	-0.004 (0.010)		
2016+ $\times$ Worried: job security	0.007 (0.005)		
Brown job $\times$ 2016+ $\times$ Worried: job security	-0.022 (0.015)		
Brown job $\times$ Worried: own economic situation		-0.011 (0.013)	
2016+ $\times$ Worried: own economic situation		0.009* (0.004)	
Brown job $\times$ 2016+ $\times$ Worried: own economic situation		0.001 (0.019)	
Brown job $\times$ Worried: migration			0.009 (0.009)
2016+ $\times$ Worried: migration			0.039*** (0.003)
Brown job $\times$ 2016+ $\times$ Worried: migration			-0.001 (0.013)
Individual FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
R ²	0.447	0.457	0.459
Observations	112,710	127,051	124,649

Notes: The table present results from interaction specifications. Prior attitudes are measured as levels in 2015. Standard errors are clustered at the individual level. Signif. Codes: ***: 0.001, **: 0.01, *: 0.05